

- g_r be the coefficients of x^r in $f(x)$ and $g(x)$, respectively (one of which is nonzero by definition of r). Then $\mathbb{Z}f_r + \mathbb{Z}g_r = \mathbb{Z}d_r$ for some nonzero $d_r \in \mathbb{Q}$ (cf. Exercise 14 in Section 2.4). Prove that there is a polynomial $d(x) \in \mathbb{Q}[x]$ that is a g.c.d. of $f(x)$ and $g(x)$ in $\mathbb{Q}[x]$ and whose term of minimal degree is $d_r x^r$.
- (b) Prove that $f(x) = d(x)q_1(x)$ and $g(x) = d(x)q_2(x)$ where $q_1(x)$ and $q_2(x)$ are elements of the subring R of $\mathbb{Q}[x]$.
 - (c) Prove that $d(x) = a(x)f(x) + b(x)g(x)$ for polynomials $a(x), b(x)$ in R . [The existence of $a(x), b(x)$ in the Euclidean Domain $\mathbb{Q}[x]$ is immediate. Use Exercise 11 in Section 2 to show that $a(x)$ and $b(x)$ can be chosen to lie in R .]
 - (d) Conclude from (a) and (b) that $Rf(x) + Rg(x) = Rd(x)$ in $\mathbb{Q}[x]$ and use this to prove that R is a Bezout Domain (cf. Exercise 7 in Section 8.2).
 - (e) Show that (d), the results of the previous exercise, and Exercise 11 of Section 8.3 imply that R must contain ideals that are not principal (hence not finitely generated). Prove that in fact $I = x\mathbb{Q}[x]$ is an ideal of R that is not finitely generated.

9.4 IRREDUCIBILITY CRITERIA

If R is a Unique Factorization Domain, then by Corollary 8 a polynomial ring in any number of variables with coefficients in R is also a Unique Factorization Domain. It is of interest then to determine the irreducible elements in such a polynomial ring, particularly in the ring $R[x]$. In the one-variable case, a nonconstant monic polynomial is irreducible in $R[x]$ if it cannot be factored as the product of two other polynomials of smaller degrees. Determining whether a polynomial has factors is frequently difficult to check, particularly for polynomials of large degree in several variables. The purpose of irreducibility criteria is to give an easier mechanism for determining when some types of polynomials are irreducible.

For the most part we restrict attention to polynomials in one variable where the coefficient ring is a Unique Factorization Domain. By Gauss' Lemma it suffices to consider factorizations in $F[x]$ where F is the field of fractions of R (although we shall occasionally consider questions of irreducibility when the coefficient ring is just an integral domain). The next proposition considers when there is a factor of degree one (a *linear* factor).

Proposition 9. Let F be a field and let $p(x) \in F[x]$. Then $p(x)$ has a factor of degree one if and only if $p(x)$ has a root in F , i.e., there is an $\alpha \in F$ with $p(\alpha) = 0$.

Proof: If $p(x)$ has a factor of degree one, then since F is a field, we may assume the factor is monic, i.e., is of the form $(x - \alpha)$ for some $\alpha \in F$. But then $p(\alpha) = 0$. Conversely, suppose $p(\alpha) = 0$. By the Division Algorithm in $F[x]$ we may write

$$p(x) = q(x)(x - \alpha) + r$$

where r is a constant. Since $p(\alpha) = 0$, r must be 0, hence $p(x)$ has $(x - \alpha)$ as a factor.

Proposition 9 gives a criterion for irreducibility for polynomials of small degree:

Proposition 10. A polynomial of degree two or three over a field F is reducible if and only if it has a root in F .

Proof: This follows immediately from the previous proposition, since a polynomial of degree two or three is reducible if and only if it has at least one linear factor.

The next result limits the possibilities for roots of polynomials with integer coefficients (it is stated for $\mathbb{Z}[x]$ for convenience although it clearly generalizes to $R[x]$, where R is any Unique Factorization Domain).

Proposition 11. Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ be a polynomial of degree n with integer coefficients. If $r/s \in \mathbb{Q}$ is in lowest terms (i.e., r and s are relatively prime integers) and r/s is a root of $p(x)$, then r divides the constant term and s divides the leading coefficient of $p(x)$: $r \mid a_0$ and $s \mid a_n$. In particular, if $p(x)$ is a *monic* polynomial with integer coefficients and $p(d) \neq 0$ for all integers d dividing the constant term of $p(x)$, then $p(x)$ has no roots in $\mathbb{Q}$.

Proof: By hypothesis, $p(r/s) = 0 = a_n(r/s)^n + a_{n-1}(r/s)^{n-1} + \cdots + a_0$. Multiplying through by s^n gives

$$0 = a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_0 s^n.$$

Thus $a_n r^n = s(-a_{n-1} r^{n-1} - \cdots - a_0 s^{n-1})$, so s divides $a_n r^n$. By assumption, s is relatively prime to r and it follows that $s \mid a_n$. Similarly, solving the equation for $a_0 s^n$ shows that $r \mid a_0$. The last assertion of the proposition follows from the previous ones.

Examples

- (1) The polynomial $x^3 - 3x - 1$ is irreducible in $\mathbb{Z}[x]$. To prove this, by Gauss' Lemma and Proposition 10 it suffices to show it has no rational roots. By Proposition 11 the only candidates for rational roots are integers which divide the constant term 1, namely ± 1 . Substituting both 1 and -1 into the polynomial shows that these are not roots.
- (2) For p any prime the polynomials $x^2 - p$ and $x^3 - p$ are irreducible in $\mathbb{Q}[x]$. This is because they have degrees ≤ 3 so it suffices to show they have no rational roots. By Proposition 11 the only candidates for roots are ± 1 and $\pm p$, but none of these give 0 when they are substituted into the polynomial.
- (3) The polynomial $x^2 + 1$ is reducible in $\mathbb{Z}/2\mathbb{Z}[x]$ since it has 1 as a root, and it factors as $(x + 1)^2$.
- (4) The polynomial $x^2 + x + 1$ is irreducible in $\mathbb{Z}/2\mathbb{Z}[x]$ since it does not have a root in $\mathbb{Z}/2\mathbb{Z}$: $0^2 + 0 + 1 = 1$ and $1^2 + 1 + 1 = 1$.
- (5) Similarly, the polynomial $x^3 + x + 1$ is irreducible in $\mathbb{Z}/2\mathbb{Z}[x]$.

This technique is limited to polynomials of low degree because it relies on the presence of a factor of degree one. A polynomial of degree 4, for example, may be the product of two irreducible quadratics, hence be reducible but have no linear factor. One fairly general technique for checking irreducibility uses Proposition 2 above and consists of reducing the coefficients modulo some ideal.

Proposition 12. Let I be a proper ideal in the integral domain R and let $p(x)$ be a nonconstant monic polynomial in $R[x]$. If the image of $p(x)$ in $(R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of smaller degree, then $p(x)$ is irreducible in $R[x]$.

Proof: Suppose $p(x)$ cannot be factored in $(R/I)[x]$ but that $p(x)$ is reducible in $R[x]$. As noted at the end of the preceding section this means there are monic, nonconstant polynomials $a(x)$ and $b(x)$ in $R[x]$ such that $p(x) = a(x)b(x)$. By Proposition 2, reducing the coefficients modulo I gives a factorization in $(R/I)[x]$ with nonconstant factors, a contradiction.

This proposition indicates that if it is possible to find a proper ideal I such that the *reduced* polynomial cannot be factored, then the polynomial is itself irreducible. Unfortunately, there are examples of polynomials even in $\mathbb{Z}[x]$ which are irreducible but whose reductions modulo every ideal are reducible (so their irreducibility is not detectable by this technique). For example, the polynomial $x^4 + 1$ is irreducible in $\mathbb{Z}[x]$ but is reducible modulo every prime (we shall verify this in Chapter 14) and the polynomial $x^4 - 72x^2 + 4$ is irreducible in $\mathbb{Z}[x]$ but is reducible modulo every integer.

Examples

- (1) Consider the polynomial $p(x) = x^2 + x + 1$ in $\mathbb{Z}[x]$. Reducing modulo 2, we see from Example 4 above that $p(x)$ is irreducible in $\mathbb{Z}[x]$. Similarly, $x^3 + x + 1$ is irreducible in $\mathbb{Z}[x]$ because it is irreducible in $\mathbb{Z}/2\mathbb{Z}[x]$.
- (2) The polynomial $x^2 + 1$ is irreducible in $\mathbb{Z}[x]$ since it is irreducible in $\mathbb{Z}/3\mathbb{Z}[x]$ (no root in $\mathbb{Z}/3\mathbb{Z}$), but is reducible mod 2. This shows that the converse to Proposition 12 does not hold.
- (3) The idea of reducing modulo an ideal to determine irreducibility can be used also in several variables, but some care must be exercised. For example, the polynomial $x^2 + xy + 1$ in $\mathbb{Z}[x, y]$ is irreducible since modulo the ideal (y) it is $x^2 + 1$ in $\mathbb{Z}[x]$, which is irreducible and of the same degree. In this sort of argument it is necessary to be careful about “collapsing.” For example, the polynomial $xy + x + y + 1$ (which is $(x + 1)(y + 1)$) is reducible, but appears irreducible modulo both (x) and (y) . The reason for this is that nonunit polynomials in $\mathbb{Z}[x, y]$ can reduce to units in the quotient. To take account of this it is necessary to determine which elements in the original ring become units in the quotient. The elements in $\mathbb{Z}[x, y]$ which are units modulo (y) , for example, are the polynomials in $\mathbb{Z}[x, y]$ with constant term ± 1 and all nonconstant terms divisible by y . The fact that $x^2 + xy + 1$ and its reduction mod (y) have the same degree therefore eliminates the possibility of a factor which is a unit modulo (y) , but not a unit in $\mathbb{Z}[x, y]$ and gives the irreducibility of this polynomial.

A special case of reducing modulo an ideal to test for irreducibility which is frequently useful is known as *Eisenstein's Criterion* (although originally proved earlier by Schönemann, so more properly known as the *Eisenstein-Schönemann Criterion*):

Proposition 13. (Eisenstein's Criterion) Let P be a prime ideal of the integral domain R and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a polynomial in $R[x]$ (here $n \geq 1$). Suppose $a_{n-1}, \dots, a_1, a_0$ are all elements of P and suppose a_0 is not an element of P^2 . Then $f(x)$ is irreducible in $R[x]$.

Proof: Suppose $f(x)$ were reducible, say $f(x) = a(x)b(x)$ in $R[x]$, where $a(x)$ and $b(x)$ are nonconstant polynomials. Reducing this equation modulo P and using the assumptions on the coefficients of $f(x)$ we obtain the equation $x^n = \overline{a(x)b(x)}$ in $(R/P)[x]$, where the bar denotes the polynomials with coefficients reduced mod P . Since P is a prime ideal, R/P is an integral domain, and it follows that both $\overline{a(x)}$ and $\overline{b(x)}$ have 0 constant term, i.e., the constant terms of both $a(x)$ and $b(x)$ are elements of P . But then the constant term a_0 of $f(x)$ as the product of these two would be an element of P^2 , a contradiction.

Eisenstein's Criterion is most frequently applied to $\mathbb{Z}[x]$ so we state the result explicitly for this case:

Corollary 14. (*Eisenstein's Criterion for $\mathbb{Z}[x]$*) Let p be a prime in $\mathbb{Z}$ and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in \mathbb{Z}[x]$, $n \geq 1$. Suppose p divides a_i for all $i \in \{0, 1, \dots, n-1\}$ but that p^2 does not divide a_0 . Then $f(x)$ is irreducible in both $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

Proof: This is simply a restatement of Proposition 13 in the case of the prime ideal (p) in $\mathbb{Z}$ together with Corollary 6.

Examples

- (1) The polynomial $x^4 + 10x + 5$ in $\mathbb{Z}[x]$ is irreducible by Eisenstein's Criterion applied for the prime 5.
- (2) If a is any integer which is divisible by some prime p but not divisible by p^2 , then $x^n - a$ is irreducible in $\mathbb{Z}[x]$ by Eisenstein's Criterion. In particular, $x^n - p$ is irreducible for all positive integers n and so for $n \geq 2$ the n^{th} roots of p are not rational numbers (i.e., this polynomial has no root in $\mathbb{Q}$).
- (3) Consider the polynomial $f(x) = x^4 + 1$ mentioned previously. Eisenstein's Criterion does not apply directly to $f(x)$. The polynomial $g(x) = f(x+1)$ is $(x+1)^4 + 1$, i.e., $x^4 + 4x^3 + 6x^2 + 4x + 2$, and Eisenstein's Criterion for the prime 2 shows that this polynomial is irreducible. It follows then that $f(x)$ must also be irreducible, since any factorization for $f(x)$ would provide a factorization for $g(x)$ (just replace x by $x+1$ in each of the factors). This example shows that Eisenstein's Criterion can sometimes be used to verify the irreducibility of a polynomial to which it does not immediately apply.
- (4) As another example of this, let p be a prime and consider the polynomial

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1,$$

an example of a *cyclotomic polynomial* which we shall consider more thoroughly in Part IV. Again, Eisenstein's Criterion does not immediately apply, but it does apply for the prime p to the polynomial

$$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \cdots + \frac{p(p-1)}{2}x + p \in \mathbb{Z}[x]$$

since all the coefficients except the first are divisible by p by the Binomial Theorem. As before, this shows $\Phi_p(x)$ is irreducible in $\mathbb{Z}[x]$.

- (5) As an example of the use of the more general Eisenstein's Criterion in Proposition 13 we mimic Example 2 above. Let $R = \mathbb{Q}[x]$ and let n be any positive integer. Consider

the polynomial $X^n - x$ in the ring $R[X]$. The ideal (x) is prime in the coefficient ring R since $R/(x) = \mathbb{Q}[x]/(x)$ is the integral domain $\mathbb{Q}$. Eisenstein's Criterion for the ideal (x) of R applies directly to show that $X^n - x$ is irreducible in $R[X]$. Note that this construction works with $\mathbb{Q}$ replaced by any field or, indeed, by any integral domain.

There are now efficient algorithms for factoring polynomials over certain fields. For polynomials with integer coefficients these algorithms have been implemented in a number of computer packages. An efficient algorithm for factoring polynomials over $\mathbb{F}_p$, called the Berlekamp Algorithm, is described in detail in the exercises at the end of Section 14.3.

EXERCISES

- Determine whether the following polynomials are irreducible in the rings indicated. For those that are reducible, determine their factorization into irreducibles. The notation $\mathbb{F}_p$ denotes the finite field $\mathbb{Z}/p\mathbb{Z}$, p a prime.
 - $x^2 + x + 1$ in $\mathbb{F}_2[x]$.
 - $x^3 + x + 1$ in $\mathbb{F}_3[x]$.
 - $x^4 + 1$ in $\mathbb{F}_5[x]$.
 - $x^4 + 10x^2 + 1$ in $\mathbb{Z}[x]$.
- Prove that the following polynomials are irreducible in $\mathbb{Z}[x]$:
 - $x^4 - 4x^3 + 6$
 - $x^6 + 30x^5 - 15x^3 + 6x - 120$
 - $x^4 + 4x^3 + 6x^2 + 2x + 1$ [Substitute $x - 1$ for x .]
 - $\frac{(x+2)^p - 2^p}{x}$, where p is an odd prime.
- Show that the polynomial $(x-1)(x-2) \cdots (x-n) - 1$ is irreducible over $\mathbb{Z}$ for all $n \geq 1$. [If the polynomial factors consider the values of the factors at $x = 1, 2, \dots, n$.]
- Show that the polynomial $(x-1)(x-2) \cdots (x-n) + 1$ is irreducible over $\mathbb{Z}$ for all $n \geq 1$, $n \neq 4$.
- Find all the monic irreducible polynomials of degree ≤ 3 in $\mathbb{F}_2[x]$, and the same in $\mathbb{F}_3[x]$.
- Construct fields of each of the following orders: (a) 9, (b) 49, (c) 8, (d) 81 (you may exhibit these as $F[x]/(f(x))$ for some F and f). [Use Exercises 2 and 3 in Section 2.]
- Prove that $\mathbb{R}[x]/(x^2 + 1)$ is a field which is isomorphic to the complex numbers.
- Prove that $K_1 = \mathbb{F}_{11}[x]/(x^2 + 1)$ and $K_2 = \mathbb{F}_{11}[y]/(y^2 + 2y + 2)$ are both fields with 121 elements. Prove that the map which sends the element $p(\bar{x})$ of K_1 to the element $p(\bar{y} + 1)$ of K_2 (where p is any polynomial with coefficients in $\mathbb{F}_{11}$) is well defined and gives a ring (hence field) isomorphism from K_1 to K_2 .
- Prove that the polynomial $x^2 - \sqrt{2}$ is irreducible over $\mathbb{Z}[\sqrt{2}]$ (you may use the fact that $\mathbb{Z}[\sqrt{2}]$ is a U.F.D. — cf. Exercise 9 of Section 8.1).
- Prove that the polynomial $p(x) = x^4 - 4x^2 + 8x + 2$ is irreducible over the quadratic field $F = \mathbb{Q}(\sqrt{-2}) = \{a + b\sqrt{-2} \mid a, b \in \mathbb{Q}\}$. [First use the method of Proposition 11 for the Unique Factorization Domain $\mathbb{Z}[\sqrt{-2}]$ (cf. Exercise 8, Section 8.1) to show that if $\alpha \in \mathbb{Z}[\sqrt{-2}]$ is a root of $p(x)$ then α is a divisor of 2 in $\mathbb{Z}[\sqrt{-2}]$. Conclude that α must be $\pm 1, \pm\sqrt{-2}$ or ± 2 , and hence show $p(x)$ has no linear factor over F . Show similarly that $p(x)$ is not the product of two quadratics with coefficients in F .]

11. Prove that $x^2 + y^2 - 1$ is irreducible in $\mathbb{Q}[x, y]$.
12. Prove that $x^{n-1} + x^{n-2} + \cdots + x + 1$ is irreducible over $\mathbb{Z}$ if and only if n is a prime.
13. Prove that $x^3 + nx + 2$ is irreducible over $\mathbb{Z}$ for all integers $n \neq 1, -3, -5$.
14. Factor each of the two polynomials: $x^8 - 1$ and $x^6 - 1$ into irreducibles over each of the following rings: (a) $\mathbb{Z}$, (b) $\mathbb{Z}/2\mathbb{Z}$, (c) $\mathbb{Z}/3\mathbb{Z}$.
15. Prove that if F is a field then the polynomial $X^n - x$ which has coefficients in the ring $F[[x]]$ of formal power series (cf. Exercise 3 of Section 7.2) is irreducible over $F[[x]]$. [Recall that $F[[x]]$ is a Euclidean Domain — cf. Exercise 5, Section 7.2 and Example 4, Section 8.1.]
16. Let F be a field and let $f(x)$ be a polynomial of degree n in $F[x]$. The polynomial $g(x) = x^n f(1/x)$ is called the *reverse* of $f(x)$.
 - (a) Describe the coefficients of g in terms of the coefficients of f .
 - (b) Prove that f is irreducible if and only if g is irreducible.
17. Prove the following variant of Eisenstein's Criterion: let P be a prime ideal in the Unique Factorization Domain R and let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial in $R[x]$, $n \geq 1$. Suppose $a_n \notin P$, $a_{n-1}, \dots, a_0 \in P$ and $a_0 \notin P^2$. Prove that $f(x)$ is irreducible in $F[x]$, where F is the quotient field of R .
18. Show that $6x^5 + 14x^3 - 21x + 35$ and $18x^5 - 30x^2 + 120x + 360$ are irreducible in $\mathbb{Q}[x]$.
19. Let F be a field and let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in F[x]$. The *derivative*, $D_x(f(x))$, of $f(x)$ is defined by

$$D_x(f(x)) = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \cdots + a_1$$

where, as usual, $na = a + a + \cdots + a$ (n times). Note that $D_x(f(x))$ is again a polynomial with coefficients in F .

The polynomial $f(x)$ is said to have a *multiple root* if there is some field E containing F and some $\alpha \in E$ such that $(x - \alpha)^2$ divides $f(x)$ in $E[x]$. For example, the polynomial $f(x) = (x - 1)^2(x - 2) \in \mathbb{Q}[x]$ has $\alpha = 1$ as a multiple root and the polynomial $f(x) = x^4 + 2x^2 + 1 = (x^2 + 1)^2 \in \mathbb{R}[x]$ has $\alpha = \pm i \in \mathbb{C}$ as multiple roots. We shall prove in Section 13.5 that a nonconstant polynomial $f(x)$ has a multiple root if and only if $f(x)$ is not relatively prime to its derivative (which can be detected by the Euclidean Algorithm in $F[x]$). Use this criterion to determine whether the following polynomials have multiple roots:

- (a) $x^3 - 3x - 2 \in \mathbb{Q}[x]$
- (b) $x^3 + 3x + 2 \in \mathbb{Q}[x]$
- (c) $x^6 - 4x^4 + 6x^3 + 4x^2 - 12x + 9 \in \mathbb{Q}[x]$
- (d) Show for any prime p and any $a \in \mathbb{F}_p$ that the polynomial $x^p - a$ has a multiple root.
20. Show that the polynomial $f(x) = x$ in $\mathbb{Z}/6\mathbb{Z}[x]$ factors as $(3x + 4)(4x + 3)$, hence is not an irreducible polynomial.
 - (a) Show that the reduction of $f(x)$ modulo both of the nontrivial ideals (2) and (3) of $\mathbb{Z}/6\mathbb{Z}$ is an irreducible polynomial, showing that the condition that R be an integral domain in Proposition 12 is necessary.
 - (b) Show that in any factorization $f(x) = g(x)h(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$ the reduction of $g(x)$ modulo (2) is either 1 or x and the reduction of $h(x)$ modulo (2) is then either x or 1, and similarly for the reductions modulo (3). Determine all the factorizations of $f(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$. [Use the Chinese Remainder Theorem.]
 - (c) Show that the ideal (3, x) is a principal ideal in $\mathbb{Z}/6\mathbb{Z}[x]$.
 - (d) Show that over the ring $\mathbb{Z}/30\mathbb{Z}[x]$ the polynomial $f(x) = x$ has the factorization

$f(x) = (10x + 21)(15x + 16)(6x + 25)$. Prove that the product of any of these factors is again of the same degree. Prove that the reduction of $f(x)$ modulo any prime in $\mathbb{Z}/30\mathbb{Z}$ is an irreducible polynomial. Determine all the factorizations of $f(x)$ in $\mathbb{Z}/30\mathbb{Z}[x]$. [Consider the reductions modulo (2), (3) and (5) and use the Chinese Remainder Theorem.]

(e) Generalize part (d) to $\mathbb{Z}/n\mathbb{Z}[x]$ where n is the product of k distinct primes.

9.5 POLYNOMIAL RINGS OVER FIELDS II

Let F be a field. We prove here some additional results for the one-variable polynomial ring $F[x]$. The first is a restatement of results obtained earlier.

Proposition 15. The maximal ideals in $F[x]$ are the ideals $(f(x))$ generated by irreducible polynomials $f(x)$. In particular, $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible.

Proof: This follows from Proposition 7 of Section 8.2 applied to the Principal Ideal Domain $F[x]$.

Proposition 16. Let $g(x)$ be a nonconstant element of $F[x]$ and let

$$g(x) = f_1(x)^{n_1} f_2(x)^{n_2} \cdots f_k(x)^{n_k}$$

be its factorization into irreducibles, where the $f_i(x)$ are distinct. Then we have the following isomorphism of rings:

$$F[x]/(g(x)) \cong F[x]/(f_1(x)^{n_1}) \times F[x]/(f_2(x)^{n_2}) \times \cdots \times F[x]/(f_k(x)^{n_k}).$$

Proof: This follows from the Chinese Remainder Theorem (Theorem 7.17), since the ideals $(f_i(x)^{n_i})$ and $(f_j(x)^{n_j})$ are comaximal if $f_i(x)$ and $f_j(x)$ are distinct (they are relatively prime in the Euclidean Domain $F[x]$, hence the ideal generated by them is $F[x]$).

The next result concerns the number of roots of a polynomial over a field F . By Proposition 9, a root α corresponds to a linear factor $(x - \alpha)$ of $f(x)$. If $f(x)$ is divisible by $(x - \alpha)^m$ but not by $(x - \alpha)^{m+1}$, then α is said to be a root of *multiplicity* m .

Proposition 17. If the polynomial $f(x)$ has roots $\alpha_1, \alpha_2, \dots, \alpha_k$ in F (not necessarily distinct), then $f(x)$ has $(x - \alpha_1) \cdots (x - \alpha_k)$ as a factor. In particular, a polynomial of degree n in one variable over a field F has at most n roots in F , even counted with multiplicity.

Proof: The first statement follows easily by induction from Proposition 9. Since linear factors are irreducible, the second statement follows since $F[x]$ is a Unique Factorization Domain.

This last result has the following interesting consequence.

Proposition 18. A finite subgroup of the multiplicative group of a field is cyclic. In particular, if F is a finite field, then the multiplicative group $F^\times$ of nonzero elements of F is a cyclic group.

Proof: We give a proof of this result using the Fundamental Theorem of Finitely Generated Abelian Groups (Theorem 3 in Section 5.2). A more number-theoretic proof is outlined in the exercises, or Proposition 5 in Section 6.1 may be used in place of the Fundamental Theorem. By the Fundamental Theorem, the finite subgroup can be written as the direct product of cyclic groups

$$\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_k \mid n_{k-1} \mid \cdots \mid n_2 \mid n_1$. In general, if G is a cyclic group and $d \mid |G|$ then G contains precisely d elements of order dividing d . Since n_k divides the order of each of the cyclic groups in the direct product, it follows that each direct factor contains n_k elements of order dividing n_k . If k were greater than 1, there would therefore be a total of more than n_k such elements. But then there would be more than n_k roots of the polynomial $x^{n_k} - 1$ in the field F , contradicting Proposition 17. Hence $k = 1$ and the group is cyclic.

Corollary 19. Let p be a prime. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero residue classes mod p is cyclic.

Proof: This is the multiplicative group of the finite field $\mathbb{Z}/p\mathbb{Z}$.

Corollary 20. Let $n \geq 2$ be an integer with factorization $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ in $\mathbb{Z}$, where $p_1, \dots, p_r$ are distinct primes. We have the following isomorphisms of (multiplicative) groups:

- (1) $(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})^\times$
- (2) $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is the direct product of a cyclic group of order 2 and a cyclic group of order $2^{\alpha-2}$, for all $\alpha \geq 2$
- (3) $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group of order $p^{\alpha-1}(p-1)$, for all odd primes p .

Remark: These isomorphisms describe the group-theoretic structure of the automorphism group of the cyclic group, Z_n , of order n since $\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ (cf. Proposition 16 in Section 4.4). In particular, for p a prime the automorphism group of the cyclic group of order p is cyclic of order $p-1$.

Proof: This is mainly a matter of collecting previous results. The isomorphism in (1) follows from the Chinese Remainder Theorem (see Corollary 18, Section 7.6). The isomorphism in (2) follows directly from Exercises 22 and 23 of Section 2.3.

For p an odd prime, $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is an abelian group of order $p^{\alpha-1}(p-1)$. By Exercise 21 of Section 2.3 the Sylow p -subgroup of this group is cyclic. The map

$$\mathbb{Z}/p^\alpha\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z} \quad \text{defined by} \quad a + (p^\alpha) \mapsto a + (p)$$

is a ring homomorphism (reduction mod p) which gives a surjective group homomorphism from $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ onto $(\mathbb{Z}/p\mathbb{Z})^\times$. The latter group is cyclic of order $p-1$

(Corollary 19). The kernel of this map is of order $p^{\alpha-1}$, hence for all primes $q \neq p$, the Sylow q -subgroup of $(\mathbb{Z}/p^{\alpha}\mathbb{Z})^{\times}$ maps isomorphically into the cyclic group $(\mathbb{Z}/p\mathbb{Z})^{\times}$. All Sylow subgroups of $(\mathbb{Z}/p^{\alpha}\mathbb{Z})^{\times}$ are therefore cyclic, so (3) holds, completing the proof.

EXERCISES

1. Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Describe the nilradical of $F[x]/(f(x))$ in terms of the factorization of $f(x)$ (cf. Exercise 29, Section 7.3).
2. For each of the fields constructed in Exercise 6 of Section 4 exhibit a generator for the (cyclic) multiplicative group of nonzero elements.
3. Let p be an odd prime in $\mathbb{Z}$ and let n be a positive integer. Prove that $x^n - p$ is irreducible over $\mathbb{Z}[i]$. [Use Proposition 18 in Chapter 8 and Eisenstein's Criterion.]
4. Prove that $x^3 + 12x^2 + 18x + 6$ is irreducible over $\mathbb{Z}[i]$. [Use Proposition 8.18 and Eisenstein's Criterion.]
5. Let φ denote Euler's φ -function. Prove the identity $\sum_{d|n} \varphi(d) = n$, where the sum is extended over all the divisors d of n . [First observe that the identity is valid when $n = p^m$ is the power of a prime p since the sum telescopes. Write $n = p^m n'$ where p does not divide n' . Prove that $\sum_{d|n} \varphi(d) = \sum_{d''|p^m} \varphi(d'') \sum_{d'|n'} \varphi(d')$ by multiplying out the right hand side and using the multiplicativity $\varphi(ab) = \varphi(a)\varphi(b)$ when a and b are relatively prime. Use induction to complete the proof. This problem may be done alternatively by letting Z be the cyclic group of order n and showing that since Z contains a unique subgroup of order d for each d dividing n , the number of elements of Z of order d is $\varphi(d)$. Then $|Z|$ is the sum of $\varphi(d)$ as d runs over all divisors of n .]
6. Let G be a finite subgroup of order n of the multiplicative group $F^{\times}$ of nonzero elements of the field F . Let φ denote Euler's φ -function and let $\psi(d)$ denote the number of elements of G of order d . Prove that $\psi(d) = \varphi(d)$ for every divisor d of n . In particular conclude that $\psi(n) \geq 1$, so that G is a cyclic group. [Observe that for any integer $N \geq 1$ the polynomial $x^N - 1$ has at most N roots in F . Conclude that for any integer N we have $\sum_{d|N} \psi(d) \leq N$. Since $\sum_{d|N} \varphi(d) = N$ by the previous exercise, show by induction that $\psi(d) \leq \varphi(d)$ for every divisor d of n . Since $\sum_{d|n} \psi(d) = n = \sum_{d|n} \varphi(d)$ show that this implies $\psi(d) = \varphi(d)$ for every divisor d of n .]
7. Prove that the additive and multiplicative groups of a field are never isomorphic. [Consider three cases: when $|F|$ is finite, when $-1 \neq 1$ in F , and when $-1 = 1$ in F .]

9.6 POLYNOMIALS IN SEVERAL VARIABLES OVER A FIELD AND GRÖBNER BASES

In this section we consider polynomials in many variables, present some basic computational tools, and indicate some applications. The results of this section are not required in Chapters 10 through 14. Additional applications will be given in Chapter 15.

We proved in Section 2 that a polynomial ring $F[x]$ in a variable x over a field F is a Euclidean Domain, and Corollary 8 showed that the polynomial ring $F[x_1, \dots, x_n]$ is a U.F.D. However it follows from Corollary 8 in Section 8.2 that the latter ring is not a P.I.D. unless $n = 1$. Our first result below shows that ideals in such polynomial rings, although not necessarily principal, are always finitely generated. General rings with this property are given a special name:

Definition. A commutative ring R with 1 is called *Noetherian* if every ideal of R is finitely generated.

Noetherian rings will be studied in greater detail in Chapters 15 and 16. In this section we develop some of the basic theory and resulting algorithms for working with (finitely generated) ideals in $F[x_1, \dots, x_n]$.

As we saw in Section 1, a polynomial ring in n variables can be considered as a polynomial ring in one variable with coefficients in a polynomial ring in $n - 1$ variables. By following this inductive approach—as we did in Theorem 7 and Corollary 8—we can deduce that $F[x_1, x_2, \dots, x_n]$ is Noetherian from the following more general result.

Theorem 21. (Hilbert's Basis Theorem) If R is a Noetherian ring then so is the polynomial ring $R[x]$.

Proof: Let I be an ideal in $R[x]$ and let L be the set of all leading coefficients of the elements in I . We first show that L is an ideal of R , as follows. Since I contains the zero polynomial, $0 \in L$. Let $f = ax^d + \dots$ and $g = bx^e + \dots$ be polynomials in I of degrees d, e and leading coefficients $a, b \in R$. Then for any $r \in R$ either $ra - b$ is zero or it is the leading coefficient of the polynomial $rx^e f - x^d g$. Since the latter polynomial is in I we have $ra - b \in L$, which shows L is an ideal of R . Since R is assumed Noetherian, the ideal L in R is finitely generated, say by $a_1, a_2, \dots, a_n \in R$. For each $i = 1, \dots, n$ let f_i be an element of I whose leading coefficient is a_i . Let e_i denote the degree of f_i , and let N be the maximum of $e_1, e_2, \dots, e_n$.

For each $d \in \{0, 1, \dots, N - 1\}$, let L_d be the set of all leading coefficients of polynomials in I of degree d together with 0. A similar argument as that for L shows each L_d is also an ideal of R , again finitely generated since R is Noetherian. For each nonzero ideal L_d let $b_{d,1}, b_{d,2}, \dots, b_{d,n_d} \in R$ be a set of generators for L_d , and let $f_{d,i}$ be a polynomial in I of degree d with leading coefficient $b_{d,i}$.

We show that the polynomials $f_1, \dots, f_n$ together with all the polynomials $f_{d,i}$ for all the nonzero ideals L_d are a set of generators for I , i.e., that

$$I = (\{f_1, \dots, f_n\} \cup \{f_{d,i} \mid 0 \leq d < N, 1 \leq i \leq n_d\}).$$

By construction, the ideal I' on the right above is contained in I since all the generators were chosen in I . If $I' \neq I$, there exists a nonzero polynomial $f \in I$ of minimum degree with $f \notin I'$. Let $d = \deg f$ and let a be the leading coefficient of f .

Suppose first that $d \geq N$. Since $a \in L$ we may write a as an R -linear combination of the generators of L : $a = r_1 a_1 + \dots + r_n a_n$. Then $g = r_1 x^{d-e_1} f_1 + \dots + r_n x^{d-e_n} f_n$ is an element of I' with the same degree d and the same leading coefficient a as f . Then $f - g \in I$ is a polynomial in I of smaller degree than f . By the minimality of f , we must have $f - g = 0$, so $f = g \in I'$, a contradiction.

Suppose next that $d < N$. In this case $a \in L_d$ for some $d < N$, and so we may write $a = r_1 b_{d,1} + \dots + r_{n_d} b_{d,n_d}$ for some $r_i \in R$. Then $g = r_1 f_{d,1} + \dots + r_{n_d} f_{d,n_d}$ is a polynomial in I' with the same degree d and the same leading coefficient a as f , and we have a contradiction as before.

It follows that $I = I'$ is finitely generated, and since I was arbitrary, this completes the proof that $R[x]$ is Noetherian.

Since a field is clearly Noetherian, Hilbert's Basis Theorem and induction immediately give:

Corollary 22. Every ideal in the polynomial ring $F[x_1, x_2, \dots, x_n]$ with coefficients from a field F is finitely generated.

If I is an ideal in $F[x_1, \dots, x_n]$ generated by a (possibly infinite) set $\mathcal{S}$ of polynomials, Corollary 22 shows that I is finitely generated, and in fact I is generated by a finite number of the polynomials from the set $\mathcal{S}$ (cf. Exercise 1).

As the proof of Hilbert's Basis Theorem shows, the collection of leading coefficients of the polynomials in an ideal I in $R[x]$ forms an extremely useful ideal in R that can be used to understand I . This suggests studying “leading terms” in $F[x_1, x_2, \dots, x_n]$ more generally (and somewhat more intrinsically). To do this we need to specify a total ordering on the monomials, since without some sort of ordering we cannot in general tell which is the “leading” term of a polynomial. We implicitly chose such an ordering in the inductive proof of Corollary 22—we first viewed a polynomial f as a polynomial in x_1 with coefficients in $R = F[x_2, \dots, x_n]$, say, then viewed its “leading coefficient” in $F[x_2, \dots, x_n]$ as a polynomial in x_2 with coefficients in $F[x_3, \dots, x_n]$, etc. This is an example of a *lexicographic* monomial ordering on the polynomial ring $F[x_1, \dots, x_n]$ which is defined by first declaring an ordering of the variables, for example $x_1 > x_2 > \dots > x_n$ and then declaring that the monomial term $Ax_1^{a_1}x_2^{a_2}\dots x_n^{a_n}$ with exponents $(a_1, a_2, \dots, a_n)$ has higher order than the monomial term $Bx_1^{b_1}x_2^{b_2}\dots x_n^{b_n}$ with exponents $(b_1, b_2, \dots, b_n)$ if the first component where the n -tuples differ has $a_i > b_i$. This is analogous to the ordering used in a dictionary (hence the name), where the letter “a” comes before “b” which in turn comes before “c”, etc., and then “aardvark” comes before “abacus” (although the ‘word’ $a^2 = aa$ comes before a in the lexicographical order). Note that the ordering is only defined up to multiplication by units (elements of $F^\times$) and that multiplying two monomials by the same nonzero monomial does not change their ordering. This can be formalized in general.

Definition. A *monomial ordering* is a well ordering “ $\geq$ ” on the set of monomials that satisfies $mm_1 \geq mm_2$ whenever $m_1 \geq m_2$ for monomials m, m_1, m_2 . Equivalently, a monomial ordering may be specified by defining a well ordering on the n -tuples $\alpha = (a_1, \dots, a_n) \in \mathbb{Z}^n$ of multidegrees of monomials $Ax_1^{a_1}\dots x_n^{a_n}$ that satisfies $\alpha + \gamma \geq \beta + \gamma$ if $\alpha \geq \beta$.

It is easy to show for any monomial ordering that $m \geq 1$ for every monomial m (cf. Exercise 2). It is not difficult to show, using Hilbert's Basis Theorem, that any total ordering on monomials which for every monomial m satisfies $m \geq 1$ and $mm_1 \geq mm_2$ whenever $m_1 \geq m_2$, is necessarily a well ordering (hence a monomial ordering)—this equivalent set of axioms for a monomial ordering may be easier to verify. For simplicity we shall limit the examples to the particularly easy and intuitive lexicographic ordering, but it is important to note that there are useful computational advantages to using other monomial orderings in practice. Some additional commonly used monomial orderings are introduced in the exercises.

As mentioned, once we have a monomial ordering we can define the leading term of a polynomial:

Definition. Fix a monomial ordering on the polynomial ring $F[x_1, x_2, \dots, x_n]$.

- (1) The *leading term* of a nonzero polynomial f in $F[x_1, x_2, \dots, x_n]$, denoted $LT(f)$, is the monomial term of maximal order in f and the leading term of $f = 0$ is 0. Define the *multidegree* of f , denoted $\partial(f)$, to be the multidegree of the leading term of f .
- (2) If I is an ideal in $F[x_1, x_2, \dots, x_n]$, the *ideal of leading terms*, denoted $LT(I)$, is the ideal generated by the leading terms of all the elements in the ideal, i.e., $LT(I) = (LT(f) \mid f \in I)$.

The leading term and the multidegree of a polynomial clearly depend on the choice of the ordering. For example $LT(2xy + y^3) = 2xy$ with multidegree $(1, 1)$ if $x > y$, but $LT(2xy + y^3) = y^3$ with multidegree $(0, 3)$ if $y > x$. In particular, the leading term of a polynomial need not be the term of largest total degree. Similarly, the ideal of leading terms $LT(I)$ of an ideal I in general depends on the ordering used. Note also that the multidegree of a polynomial satisfies $\partial(fg) = \partial f + \partial g$ when f and g are nonzero, and that in this case $LT(fg) = LT(f) + LT(g)$ (cf. Exercise 2).

The ideal $LT(I)$ is by definition generated by monomials. Such ideals are called *monomial ideals* and are typically much easier to work with than generic ideals. For example, a polynomial is contained in a monomial ideal if and only if each of its monomial terms is a multiple of one of the generators for the ideal (cf. Exercise 10).

It was important in the proof of Hilbert's Basis Theorem to have *all* of the leading terms of the ideal I . If $I = (f_1, \dots, f_m)$, then $LT(I)$ contains the leading terms $LT(f_1), \dots, LT(f_m)$ of the generators for I by definition. Since $LT(I)$ is an ideal, it contains the ideal generated by these leading terms:

$$(LT(f_1), \dots, LT(f_m)) \subseteq LT(I).$$

The first of the following examples shows that the ideal $LT(I)$ of leading terms can in general be strictly larger than the ideal generated just by the leading terms of some generators for I .

Examples

- (1) Choose the lexicographic ordering $x > y$ on $F[x, y]$. The leading terms of the polynomials $f_1 = x^3y - xy^2 + 1$ and $f_2 = x^2y^2 - y^3 - 1$ are $LT(f_1) = x^3y$ (so the multidegree of f_1 is $\partial(f_1) = (3, 1)$) and $LT(f_2) = x^2y^2$ (so $\partial(f_2) = (2, 2)$). If $I = (f_1, f_2)$ is the ideal generated by f_1 and f_2 then the leading term ideal $LT(I)$ contains $LT(f_1) = x^3y$ and $LT(f_2) = x^2y^2$, so $(x^3y, x^2y^2) \subseteq LT(I)$. Since

$$yf_1 - xf_2 = y(x^3y - xy^2 + 1) - x(x^2y^2 - y^3 - 1) = x + y$$

we see that $g = x + y$ is an element of I and so the ideal $LT(I)$ also contains the leading term $LT(g) = x$. This shows that $LT(I)$ is strictly larger than $(LT(f_1), LT(f_2))$, since every element in $(LT(f_1), LT(f_2)) = (x^3y, x^2y^2)$ has total degree at least 4. We shall see later that in this case $LT(I) = (x, y^4)$.

- (2) With respect to the lexicographic ordering $y > x$, the leading terms of f_1 and f_2 in the previous example are $LT(f_1) = -xy^2$ (which one could write as $-y^2x$ to emphasize the chosen ordering) and $LT(f_2) = -y^3$. We shall see later that in this ordering $LT(I) = (x^4, y)$, which is a different ideal than the ideal $LT(I)$ obtained in the previous example using the ordering $x > y$, and is again strictly larger than $(LT(f_1), LT(f_2))$.
- (3) Choose any ordering on $F[x, y]$ and let $f = f(x, y)$ be any nonzero polynomial. The leading term of every element of the principal ideal $I = (f)$ is then a multiple of the leading term of f , so in this case $LT(I) = (LT(f))$.

In the case of one variable, leading terms are used in the Division Algorithm to reduce one polynomial g modulo another polynomial f to get a unique remainder r , and this remainder is 0 if and only if g is contained in the ideal (f) . Since $F[x_1, x_2, \dots, x_n]$ is not a Euclidean Domain if $n \geq 2$ (since it is not a P.I.D.), the situation is more complicated for polynomials in more than one variable. In the first example above, neither f_1 nor f_2 divides g in $F[x, y]$ (by degree considerations, for example), so attempting to first divide g by one of f_1 or f_2 and then by the other to try to reduce g modulo the ideal I would produce a (nonzero) “remainder” of g itself. In particular, this would suggest that $g = yf_1 - xf_2$ is not an element of the ideal I even though it is. The reason the polynomial g of degree 1 can be a linear combination of the two polynomials f_1 and f_2 of degree 4 is that the leading terms in yf_1 and xf_2 cancel in the difference, and this is reflected in the fact that $LT(f_1)$ and $LT(f_2)$ are not sufficient to generate $LT(I)$. A set of generators for an ideal I in $F[x_1, \dots, x_n]$ whose leading terms generate the leading terms of *all* the elements in I is given a special name.

Definition. A *Gröbner basis* for an ideal I in the polynomial ring $F[x_1, \dots, x_n]$ is a finite set of generators $\{g_1, \dots, g_m\}$ for I whose leading terms generate the ideal of all leading terms in I , i.e.,

$$I = (g_1, \dots, g_m) \quad \text{and} \quad LT(I) = (LT(g_1), \dots, LT(g_m)).$$

Remark: Note that a Gröbner “basis” is in fact a set of *generators* for I (that depends on the choice of ordering), i.e., every element in I is a linear combination of the generators, and not a basis in the sense of vector spaces (where the linear combination would be *unique*, cf. Sections 10.3 and 11.1). Although potentially misleading, the terminology “Gröbner basis” has been so widely adopted that it would be hazardous to introduce a different nomenclature.

One of the most important properties of a Gröbner basis (proved in Theorem 23 following) is that every polynomial g can be written *uniquely* as the sum of an element in I and a remainder r obtained by a general polynomial division. In particular, we shall see that g is an element of I if and only if this remainder r is 0. While there is a similar decomposition in general, we shall see that if we do not use a Gröbner basis the uniqueness is lost (and we cannot detect membership in I by checking whether the remainder is 0) because there are leading terms not accounted for by the leading terms of the generators.

We first use the leading terms of polynomials defined by a monomial ordering on $F[x_1, \dots, x_n]$ to extend the one variable Division Algorithm to a noncanonical polynomial division in several variables. Recall that for polynomials in one variable, the usual Division Algorithm determines the quotient $q(x)$ and remainder $r(x)$ in the equation $f(x) = q(x)g(x) + r(x)$ by successively testing whether the leading term of the dividend $f(x)$ is divisible by the leading term of $g(x)$: if $LT(f) = a(x)LT(g)$, the monomial term $a(x)$ is added to the quotient and the process is iterated with $f(x)$ replaced by the dividend $f(x) - a(x)g(x)$, which is of smaller degree since the leading terms cancel (by the choice of $a(x)$). The process terminates when the leading term of the divisor $g(x)$ no longer divides the leading term of the dividend, leaving the remainder $r(x)$. We can extend this to division by a finite number of polynomials in several variables simply by allowing successive divisions, resulting in a remainder and several quotients, as follows.

General Polynomial Division

Fix a monomial ordering on $F[x_1, \dots, x_n]$, and suppose $g_1, \dots, g_m$ is a set of nonzero polynomials in $F[x_1, \dots, x_n]$. If f is any polynomial in $F[x_1, \dots, x_n]$, start with a set of quotients $q_1, \dots, q_m$ and a remainder r initially all equal to 0 and successively test whether the leading term of the dividend f is divisible by the leading terms of the divisors $g_1, \dots, g_m$, in that order. Then

- i. If $LT(f)$ is divisible by $LT(g_i)$, say, $LT(f) = a_i LT(g_i)$, add a_i to the quotient q_i , replace f by the dividend $f - a_i g_i$ (a polynomial with lower order leading term), and reiterate the entire process.
- ii. If the leading term of the dividend f is not divisible by any of the leading terms $LT(g_1), \dots, LT(g_m)$, add the leading term of f to the remainder r , replace f by the dividend $f - LT(f)$ (i.e., remove the leading term of f), and reiterate the entire process.

The process terminates (cf. Exercise 3) when the dividend is 0 and results in a set of quotients $q_1, \dots, q_m$ and a remainder r with

$$f = q_1 g_1 + \dots + q_m g_m + r.$$

Each $q_i g_i$ has multidegree less than or equal to the multidegree of f and the remainder r has the property that no nonzero term in r is divisible by any of the leading terms $LT(g_1), \dots, LT(g_m)$ (since only terms with this property are added to r in (ii)).

Examples

Fix the lexicographic ordering $x > y$ on $F[x, y]$.

- (I) Suppose $f = x^3 y^3 + 3x^2 y^4$ and $g = xy^4$. The leading term of f is $x^3 y^3$, which is not divisible by (the leading term of) g , so $x^3 y^3$ is added to the remainder r (so now $r = x^3 y^3$) and f is replaced by $f - LT(f) = 3x^2 y^4$ and we start over. Since $3x^2 y^4$ is divisible by $LT(g) = xy^4$, with quotient $a = 3x$, we add $3x$ to the quotient q (so $q = 3x$), and replace $3x^2 y^4$ by $3x^2 y^4 - aLT(g) = 0$, at which point the process terminates. The result is the quotient $q = 3x$ and remainder $r = x^3 y^3$ and

$$x^3 y^3 + 3x^2 y^4 = f = qg + r = (3x)(xy^4) + x^3 y^3.$$

Note that if we had terminated at the first step because the leading term of f is not divisible by the leading term of g (which terminates the Division Algorithm for polynomials in one variable), then we would have been left with a 'remainder' of f itself, even though 'more' of f is divisible by g . This is the reason for step 2 in the division process (which is not necessary for polynomials in one variable).

- (2) Let $f = x^2 + x - y^2 + y$, and suppose $g_1 = xy + 1$ and $g_2 = x + y$. In the first iteration, the leading term x^2 of f is not divisible by the leading term of g_1 , but is divisible by the leading term of g_2 , so the quotient q_2 is x and the dividend f is replaced by the dividend $f - xg_2 = -xy + x - y^2 + y$. In the second iteration, the leading term of $-xy + x - y^2 + y$ is divisible by $LT(g_1)$, with quotient -1 , so $q_1 = -1$ and the dividend is replaced by $(-xy + x - y^2 + y) - (-1)g_1 = x - y^2 + y + 1$. In the third iteration, the leading term of $x - y^2 + y + 1$ is not divisible by the leading term of g_1 , but is divisible by the leading term of g_2 , with quotient 1, so 1 is added to q_2 (which is now $q_2 = x + 1$) and the dividend becomes $(x - y^2 + y + 1) - (1)g_2 = -y^2 + 1$. The leading term is now $-y^2$, which is not divisible by either $LT(g_1) = xy$ or $LT(g_2) = x$, so $-y^2$ is added to the remainder r (which is now $-y^2$) and the dividend becomes simply 1. Finally, 1 is not divisible by either $LT(g_1)$ or $LT(g_2)$, so is added to the remainder (so r is now $-y^2 + 1$), and the process terminates. The result is

$$q_1 = -1, \quad q_2 = x + 1, \quad r = -y^2 + 1 \quad \text{and}$$

$$\begin{aligned} f &= x^2 + x - y^2 + y = (-1)(xy + 1) + (x + 1)(x + y) + (-y^2 + 1) \\ &= q_1 g_1 + q_2 g_2 + r. \end{aligned}$$

- (3) Let $f = x^2 + x - y^2 + y$ as in the previous example and interchange the divisors g_1 and g_2 : $g_1 = x + y$ and $g_2 = xy + 1$. In this case an easy computation gives

$$q_1 = x - y + 1, \quad q_2 = 0, \quad r = 0 \quad \text{and}$$

$$f = x^2 + x - y^2 + y = (x - y + 1)(x + y) = q_1 g_1 + q_2 g_2 + r,$$

showing that the quotients q_i and the remainder r are in general not unique and depend on the order of the divisors $g_1, \dots, g_m$.

The computation in Example 3 shows that the polynomial $f = x^2 + x - y^2 + y$ is an element of the ideal $I = (x + y, xy + 1)$ since the remainder obtained in this case was 0 (in fact f is just a multiple of the first generator). In Example 2, however, the same polynomial resulted in a nonzero remainder $-y^2 + 1$ when divided by $xy + 1$ and $x + y$, and it was not at all clear from that computation that f was an element of I .

The next theorem shows that if we use a Gröbner basis for the ideal I then these difficulties do not arise: we obtain a *unique* remainder, which in turn can be used to determine whether a polynomial f is an element of the ideal I .

Theorem 23. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$ and suppose $\{g_1, \dots, g_m\}$ is a Gröbner basis for the nonzero ideal I in R . Then

- (1) Every polynomial $f \in R$ can be written uniquely in the form

$$f = f_I + r$$

where $f_I \in I$ and no nonzero monomial term of the 'remainder' r is divisible by any of the leading terms $LT(g_1), \dots, LT(g_m)$.

- (2) Both f_I and r can be computed by general polynomial division by $g_1, \dots, g_m$ and are independent of the order in which these polynomials are used in the division.
- (3) The remainder r provides a unique representative for the coset of f in the quotient ring $F[x_1, \dots, x_n]/I$. In particular, $f \in I$ if and only if $r = 0$.

Proof: Letting $f_I = \sum_{i=1}^m q_i g_i \in I$ in the general polynomial division of f by $g_1, \dots, g_m$ immediately gives a decomposition $f = f_I + r$ for any generators $g_1, \dots, g_m$. Suppose now that $\{g_1, \dots, g_m\}$ is a Gröbner basis, and $f = f_I + r = f'_I + r'$. Then $r - r' = f'_I - f_I \in I$, so its leading term $LT(r - r')$ is an element of $LT(I)$, which is the ideal $(LT(g_1), \dots, LT(g_m))$ since $\{g_1, \dots, g_m\}$ is a Gröbner basis for I . Every element in this ideal is a sum of multiples of the monomial terms $LT(g_1), \dots, LT(g_m)$, so is a sum of terms each of which is divisible by one of the $LT(g_i)$. But both r and r' , hence also $r - r'$, are sums of monomial terms none of which is divisible by $LT(g_1), \dots, LT(g_m)$, which is a contradiction unless $r - r' = 0$. It follows that $r = r'$ is unique, hence so is $f_I = f - r$, which proves (1).

We have already seen that f_I and r can be computed algorithmically by polynomial division, and the uniqueness in (1) implies that r is independent of the order in which the polynomials $g_1, \dots, g_m$ are used in the division. Similarly $f_I = \sum_{i=1}^m q_i g_i$ is uniquely determined (even though the individual quotients q_i are not in general unique), which gives (2).

The first statement in (3) is immediate from the uniqueness in (1). If $r = 0$, then $f = f_I \in I$. Conversely, if $f \in I$, then $f = f + 0$ together with the uniqueness of r implies that $r = 0$, and the final statement of the theorem follows.

As previously mentioned, the importance of Theorem 23, and one of the principal uses of Gröbner bases, is the uniqueness of the representative r , which allows effective computation in the quotient ring $F[x_1, \dots, x_n]/I$.

We next prove that a set of polynomials in an ideal whose leading terms generate all the leading terms of an ideal is in fact a set of generators for the ideal itself (and so is a Gröbner basis—in some works this is taken as the definition of a Gröbner basis), and this shows in particular that a Gröbner basis always exists.

Proposition 24. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$ and let I be a nonzero ideal in R .

- (1) If $g_1, \dots, g_m$ are any elements of I such that $LT(I) = (LT(g_1), \dots, LT(g_m))$, then $\{g_1, \dots, g_m\}$ is a Gröbner basis for I .
- (2) The ideal I has a Gröbner basis.

Proof: Suppose $g_1, \dots, g_m \in I$ with $LT(I) = (LT(g_1), \dots, LT(g_m))$. We need to see that $g_1, \dots, g_m$ generate the ideal I . If $f \in I$, use general polynomial division to write $f = \sum_{i=1}^m q_i g_i + r$ where no nonzero term in the remainder r is divisible by any $LT(g_i)$. Since $f \in I$, also $r \in I$, which means $LT(r)$ is in $LT(I)$. But then $LT(r)$ would be divisible by one of $LT(g_1), \dots, LT(g_m)$, which is a contradiction unless $r = 0$. Hence $f = \sum_{i=1}^m q_i g_i$ and $g_1, \dots, g_m$ generate I , so are a Gröbner basis for I , which proves (1).

For (2), note that the ideal $LT(I)$ of leading terms of any ideal I is a monomial ideal generated by all the leading terms of the polynomials in I . By Exercise 1 a finite number of those leading terms suffice to generate $LT(I)$, say $LT(I) = (LT(h_1), \dots, LT(h_k))$ for some $h_1, \dots, h_k \in I$. By (1), the polynomials $h_1, \dots, h_k$ are a Gröbner basis of I , completing the proof.

Proposition 24 proves that Gröbner bases always exist. We next prove a criterion that determines whether a given set of generators of an ideal I is a Gröbner basis, which we then use to provide an algorithm to find a Gröbner basis. The basic idea is very simple: additional elements in $LT(I)$ can arise by taking linear combinations of generators that cancel leading terms, as we saw in taking $yf_1 - xf_2$ in the first example in this section. We shall see that obtaining new leading terms from generators in this simple manner is the only obstruction to a set of generators being a Gröbner basis.

In general, if f_1, f_2 are two polynomials in $F[x_1, \dots, x_n]$ and M is the monic least common multiple of the monomial terms $LT(f_1)$ and $LT(f_2)$ then we can cancel the leading terms by taking the difference

$$S(f_1, f_2) = \frac{M}{LT(f_1)} f_1 - \frac{M}{LT(f_2)} f_2. \quad (9.1)$$

The next lemma shows that these elementary linear combinations account for all cancellation in leading terms of polynomials of the same multidegree.

Lemma 25. Suppose $f_1, \dots, f_m \in F[x_1, \dots, x_n]$ are polynomials with the same multidegree α and that the linear combination $h = a_1 f_1 + \dots + a_m f_m$ with constants $a_i \in F$ has strictly smaller multidegree. Then

$$h = \sum_{i=2}^m b_i S(f_{i-1}, f_i), \quad \text{for some constants } b_i \in F.$$

Proof: Write $f_i = c_i f'_i$ where $c_i \in F$ and f'_i is a monic polynomial of multidegree α . We have

$$\begin{aligned} h = \sum a_i c_i f'_i &= a_1 c_1 (f'_1 - f'_2) + (a_1 c_1 + a_2 c_2) (f'_2 - f'_3) + \dots \\ &\quad + (a_1 c_1 + \dots + a_{m-1} c_{m-1}) (f'_{m-1} - f'_m) + (a_1 c_1 + \dots + a_m c_m) f'_m. \end{aligned}$$

Note that $f'_{i-1} - f'_i = S(f_{i-1}, f_i)$. Then since h and each $f'_{i-1} - f'_i$ has multidegree strictly smaller than α , we have $a_1 c_1 + \dots + a_m c_m = 0$, so the last term on the right hand side is 0 and the lemma follows.

The next proposition shows that a set of generators $g_1, \dots, g_m$ is a Gröbner basis if there are no new leading terms among the differences $S(g_i, g_j)$ not already accounted for by the g_i . This result provides the principal ingredient in an algorithm to construct a Gröbner basis.

For a fixed monomial ordering on $R = F[x_1, \dots, x_n]$ and ordered set of polynomials $G = \{g_1, \dots, g_m\}$ in R , write $f \equiv r \bmod G$ if r is the remainder obtained by general polynomial division of $f \in R$ by $g_1, \dots, g_m$ (in that order).

Proposition 26. (*Buchberger's Criterion*) Let $R = F[x_1, \dots, x_n]$ and fix a monomial ordering on R . If $I = (g_1, \dots, g_m)$ is a nonzero ideal in R , then $G = \{g_1, \dots, g_m\}$ is a Gröbner basis for I if and only if $S(g_i, g_j) \equiv 0 \pmod{G}$ for $1 \leq i < j \leq m$.

Proof: If $\{g_1, \dots, g_m\}$ is a Gröbner basis for I , then $S(g_i, g_j) \equiv 0 \pmod{G}$ by Theorem 23 since each $S(g_i, g_j)$ is an element of I .

Suppose now that $S(g_i, g_j) \equiv 0 \pmod{G}$ for $1 \leq i < j \leq m$ and take any element $f \in I$. To see that G is a Gröbner basis we need to see that $(LT(g_1), \dots, LT(g_m))$ contains $LT(f)$. Since $f \in I$, we can write $f = \sum_{i=1}^m h_i g_i$ for some polynomials $h_1, \dots, h_m$. Such a representation is not unique. Among all such representations choose one for which the largest multidegree of any summand (i.e., $\max_{i=1, \dots, m} \partial(h_i g_i)$) is minimal, say α . It is clear that the multidegree of f is no worse than the largest multidegree of all the summands $h_i g_i$, so $\partial(f) \leq \alpha$. Write

$$\begin{aligned} f &= \sum_{i=1}^m h_i g_i = \sum_{\partial(h_i g_i) = \alpha} h_i g_i + \sum_{\partial(h_i g_i) < \alpha} h_i g_i \\ &= \sum_{\partial(h_i g_i) = \alpha} LT(h_i) g_i + \sum_{\partial(h_i g_i) = \alpha} (h_i - LT(h_i)) g_i + \sum_{\partial(h_i g_i) < \alpha} h_i g_i. \end{aligned} \quad (9.2)$$

Suppose that $\partial(f) < \alpha$. Then since the multidegree of the second two sums is also strictly smaller than α it follows that the multidegree of the first sum is strictly smaller than α . If $a_i \in F$ denotes the constant coefficient of the monomial term $LT(h_i)$ then $LT(h_i) = a_i h'_i$ where h'_i is a monomial. We can apply Lemma 25 to $\sum a_i (h'_i g_i)$ to write the first sum above as $\sum b_i S(h'_{i-1} g_{i-1}, h'_i g_i)$ with $\partial(h'_{i-1} g_{i-1}) = \partial(h'_i g_i) = \alpha$. Let $\beta_{i-1, i}$ be the multidegree of the monic least common multiple of $LT(g_{i-1})$ and $LT(g_i)$. Then an easy computation shows that $S(h'_{i-1} g_{i-1}, h'_i g_i)$ is just $S(g_{i-1}, g_i)$ multiplied by the monomial of multidegree $\alpha - \beta_{i-1, i}$. The polynomial $S(g_{i-1}, g_i)$ has multidegree less than $\beta_{i-1, i}$ and, by assumption, $S(g_{i-1}, g_i) \equiv 0 \pmod{G}$. This means that after general polynomial division of $S(g_{i-1}, g_i)$ by $g_1, \dots, g_m$, each $S(g_{i-1}, g_i)$ can be written as a sum $\sum q_j g_j$ with $\partial(q_j g_j) < \beta_{i-1, i}$. It follows that each $S(h'_{i-1} g_{i-1}, h'_i g_i)$ is a sum $\sum q'_j g_j$ with $\partial(q'_j g_j) < \alpha$. But then all the sums on the right hand side of equation (2) can be written as a sum of terms of the form $p_i g_i$ with polynomials p_i satisfying $\partial(p_i g_i) < \alpha$. This contradicts the minimality of α and shows that in fact $\partial(f) = \alpha$, i.e., the leading term of f has multidegree α .

If we now take the terms in equation (2) of multidegree α we see that

$$LT(f) = \sum_{\partial(h_i g_i) = \alpha} LT(h_i) LT(g_i),$$

so indeed $LT(f) \in (LT(g_1), \dots, LT(g_m))$. It follows that $G = \{g_1, \dots, g_m\}$ is a Gröbner basis.

Buchberger's Algorithm

Buchberger's Criterion can be used to provide an algorithm to find a Gröbner basis for an ideal I , as follows. If $I = (g_1, \dots, g_m)$ and each $S(g_i, g_j)$ leaves a remainder of 0 when divided by $G = \{g_1, \dots, g_m\}$ using general polynomial division then G

is a Gröbner basis. Otherwise $S(g_i, g_j)$ has a nonzero remainder r . Increase G by appending the polynomial $g_{m+1} = r$: $G' = \{g_1, \dots, g_m, g_{m+1}\}$ and begin again (note that this is again a set of generators for I since $g_{m+1} \in I$). It is not hard to check that this procedure terminates after a finite number of steps in a generating set G that satisfies Buchberger's Criterion, hence is a Gröbner basis for I (cf. Exercise 16). Note that once an $S(g_i, g_j)$ yields a remainder of 0 after division by the polynomials in G it also yields a remainder of 0 when additional polynomials are appended to G .

If $\{g_1, \dots, g_m\}$ is a Gröbner basis for the ideal I and $LT(g_j)$ is divisible by $LT(g_i)$ for some $j \neq i$, then $LT(g_j)$ is not needed as a generator for $LT(I)$. By Proposition 24 we may therefore delete g_j and still retain a Gröbner basis for I . We may also assume without loss that the leading term of each g_i is monic. A Gröbner basis $\{g_1, \dots, g_m\}$ for I where each $LT(g_i)$ is monic and where $LT(g_j)$ is not divisible by $LT(g_i)$ for $i \neq j$ is called a *minimal Gröbner basis*. While a minimal Gröbner basis is not unique, the number of elements and their leading terms are unique (cf. Exercise 15).

Examples

- (1) Choose the lexicographic ordering $x > y$ on $F[x, y]$ and consider the ideal I generated by $f_1 = x^3y - xy^2 + 1$ and $f_2 = x^2y^2 - y^3 - 1$ as in Example 1 at the beginning of this section. To test whether $G = \{f_1, f_2\}$ is a Gröbner basis we compute $S(f_1, f_2) = yf_1 - xf_2 = x + y$, which is its own remainder when divided by $\{f_1, f_2\}$, so G is not a Gröbner basis for I . Set $f_3 = x + y$, and increase the generating set: $G' = \{f_1, f_2, f_3\}$. Now $S(f_1, f_2) \equiv 0 \bmod G'$, and a brief computation yields

$$S(f_1, f_3) = f_1 - x^2yf_3 = -x^2y^2 - xy^2 + 1 \equiv 0 \bmod G'$$

$$S(f_2, f_3) = f_2 - xy^2f_3 = -xy^3 - y^3 - 1 \equiv y^4 - y^3 - 1 \bmod G'.$$

Let $f_4 = y^4 - y^3 - 1$ and increase the generating set to $G'' = \{f_1, f_2, f_3, f_4\}$. The previous 0 remainder is still 0, and now $S(f_2, f_3) \equiv 0 \bmod G''$ by the choice of f_4 . Some additional computation yields

$$S(f_1, f_4) \equiv S(f_2, f_4) \equiv S(f_3, f_4) \equiv 0 \bmod G''$$

and so $\{x^3y - xy^2 + 1, x^2y^2 - y^3 - 1, x + y, y^4 - y^3 - 1\}$ is a Gröbner basis for I . In particular, $LT(I)$ is generated by the leading terms of these four polynomials, so $LT(I) = (x^3y, x^2y^2, x, y^4) = (x, y^4)$, as previously mentioned. Then $x + y$ and $y^4 - y^3 - 1$ in I have leading terms generating $LT(I)$, so by Proposition 24, $\{x + y, y^4 - y^3 - 1\}$ gives a minimal Gröbner basis for I :

$$I = (x + y, y^4 - y^3 - 1).$$

This description of I is much simpler than $I = (x^3y - xy^2 + 1, x^2y^2 - y^3 - 1)$.

- (2) Choose the lexicographic ordering $y > x$ on $F[x, y]$ and consider the ideal I in the previous example. In this case, $S(f_1, f_2)$ produces a remainder of $f_3 = -x - y$; then $S(f_1, f_3)$ produces a remainder of $f_4 = -x^4 - x^3 + 1$, and then all remainders are 0 with respect to the Gröbner basis $\{x^3y - xy^2 + 1, x^2y^2 - y^3 - 1, -x - y, -x^4 - x^3 + 1\}$. Here $LT(I) = (-xy^2, -y^3, -y, -x^4) = (y, x^4)$, as previously mentioned, and $\{x + y, x^4 + x^3 - 1\}$ gives a minimal Gröbner basis for I with respect to this ordering:

$$I = (x + y, x^4 + x^3 - 1),$$

a different simpler description of I .

In Example 1 above it is easy to check that $\{x + y^4 - y^3 + y - 1, y^4 - y^3 - 1\}$ is again a minimal Gröbner basis for I (this is just $\{f_3 + f_4, f_4\}$), so even with a fixed monomial ordering on $F[x_1, \dots, x_n]$ a minimal Gröbner basis for an ideal I is not unique. We can obtain an important uniqueness property by strengthening the condition on divisibility by the leading terms of the basis.

Definition. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$. A Gröbner basis $\{g_1, \dots, g_m\}$ for the nonzero ideal I in R is called a *reduced Gröbner basis* if

- (a) each g_i has monic leading term, i.e., $LT(g_i)$ is monic, $i = 1, \dots, m$, and
- (b) no term in g_j is divisible by $LT(g_i)$ for $j \neq i$.

Note that a reduced Gröbner basis is, in particular, a minimal Gröbner basis. If $G = \{g_1, \dots, g_m\}$ is a minimal Gröbner basis for I , then the leading term $LT(g_j)$ is not divisible by $LT(g_i)$ for any $i \neq j$. As a result, if we use polynomial division to divide g_j by the other polynomials in G we obtain a remainder g'_j in the ideal I with the same leading term as g_j (the remainder g'_j does not depend on the order of the polynomials used in the division by (2) of Theorem 23). By Proposition 24, replacing g_j by g'_j in G again gives a minimal Gröbner basis for I , and in this basis no term of g'_j is divisible by $LT(g_i)$ for any $i \neq j$. Replacing each element in G by its remainder after division by the other elements in G therefore results in a reduced Gröbner basis for I . The importance of reduced Gröbner bases is that they are unique (for a given monomial ordering), as the next result shows.

Theorem 27. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$. Then there is a unique reduced Gröbner basis for every nonzero ideal I in R .

Proof: By Exercise 15, two reduced bases have the same number of elements and the same leading terms since reduced bases are also minimal bases. If $G = \{g_1, \dots, g_m\}$ and $G' = \{g'_1, \dots, g'_m\}$ are two reduced bases for the same nonzero ideal I , then after a possible rearrangement we may assume $LT(g_i) = LT(g'_i) = h_i$ for $i = 1, \dots, m$. For any fixed i , consider the polynomial $f_i = g_i - g'_i$. If f_i is nonzero, then since $f_i \in I$, its leading term must be divisible by some h_j . By definition of a reduced basis, h_j for $j \neq i$ does not divide any of the terms in either g_i or g'_i , hence does not divide $LT(f_i)$. But h_i also does not divide $LT(f_i)$ since all the terms in f_i have strictly smaller multidegree. This forces $f_i = 0$, i.e., $g_i = g'_i$ for every i , so $G = G'$.

One application of the uniqueness of the reduced Gröbner basis is a computational method to determine when two ideals in a polynomial ring are equal.

Corollary 28. Let I and J be two ideals in $F[x_1, \dots, x_n]$. Then $I = J$ if and only if I and J have the same reduced Gröbner basis with respect to any fixed monomial ordering on $F[x_1, \dots, x_n]$.

Examples

- (1) Consider the ideal $I = (h_1, h_2, h_3)$ with $h_1 = x^2 + xy^5 + y^4$, $h_2 = xy^6 - xy^3 + y^5 - y^2$, and $h_3 = xy^5 - xy^2$ in $F[x, y]$. Using the lexicographic ordering $x > y$ we find

$S(h_1, h_2) \equiv S(h_1, h_3) \equiv 0 \pmod{\{h_1, h_2, h_3\}}$ and $S(h_2, h_3) \equiv y^5 - y^2 \pmod{\{h_1, h_2, h_3\}}$. Setting $h_4 = y^5 - y^2$ we find $S(h_i, h_j) \equiv 0 \pmod{\{h_1, h_2, h_3, h_4\}}$ for $1 \leq i < j \leq 4$, so

$$x^2 + xy^5 + y^4, \quad xy^6 - xy^3 + y^5 - y^2, \quad xy^5 - xy^2, \quad y^5 - y^2$$

is a Gröbner basis for I . The leading terms of this basis are x^2, xy^6, xy^5, y^5 . Since y^5 divides both xy^6 and xy^5 , we may remove the second and third generators to obtain a minimal Gröbner basis $\{x^2 + xy^5 + y^4, y^5 - y^2\}$ for I . The second term in the first generator is divisible by the leading term y^5 of the second generator, so this is not a reduced Gröbner basis. Replacing $x^2 + xy^5 + y^4$ by its remainder $x^2 + xy^2 + y^4$ after division by the other polynomials in the basis (which in this case is only the polynomial $y^5 - y^2$), we are left with the reduced Gröbner basis $\{x^2 + xy^2 + y^4, y^5 - y^2\}$ for I .

- (2) Consider the ideal $J = (h_1, h_2, h_3)$ with $h_1 = xy^3 + y^3 + 1$, $h_2 = x^3y - x^3 + 1$, and $h_3 = x + y$ in $F[x, y]$. Using the lexicographic monomial ordering $x > y$ we find $S(h_1, h_2) \equiv 0 \pmod{\{h_1, h_2, h_3\}}$ and $S(h_1, h_3) \equiv y^4 - y^3 - 1 \pmod{\{h_1, h_2, h_3\}}$. Setting $h_4 = y^4 - y^3 - 1$ we find $S(h_i, h_j) \equiv 0 \pmod{\{h_1, h_2, h_3, h_4\}}$ for $1 \leq i < j \leq 4$, so

$$xy^3 + y^3 + 1, \quad x^3y - x^3 + 1, \quad x + y, \quad y^4 - y^3 - 1$$

is a Gröbner basis for J . The leading terms of this basis are xy^3, x^3y, x , and y^4 , so $\{x + y, y^4 - y^3 - 1\}$ is a minimal Gröbner basis for J . In this case none of the terms in $y^4 - y^3 - 1$ are divisible by the leading term of $x + y$ and none of the terms in $x + y$ are divisible by the leading term in $y^4 - y^3 - 1$, so $\{x + y, y^4 - y^3 - 1\}$ is the reduced Gröbner basis for J . This is the basis for the ideal I in Example 1 following Proposition 26, so these two ideals are equal:

$$(x^3y - xy^2 + 1, x^2y^2 - y^3 - 1) = (xy^3 + y^3 + 1, x^3y - x^3 + 1, x + y)$$

(and both are equal to the ideal $(x + y, y^4 - y^3 - 1)$).

Gröbner Bases and Solving Algebraic Equations: Elimination

The theory of Gröbner bases is very useful in explicitly solving systems of algebraic equations, and is the basis by which computer algebra programs attempt to solve systems of equations. Suppose $S = \{f_1, \dots, f_m\}$ is a collection of polynomials in n variables $x_1, \dots, x_n$ and we are trying to find the solutions of the system of equations $f_1 = 0$, $f_2 = 0$, $\dots$, $f_m = 0$ (i.e., the common set of zeros of the polynomials in S). If $(a_1, \dots, a_n)$ is any solution to this system, then every element f of the ideal I generated by S also satisfies $f(a_1, \dots, a_n) = 0$. Furthermore, it is an easy exercise to see that if $S' = \{g_1, \dots, g_s\}$ is any set of generators for the ideal I then the set of solutions to the system $g_1 = 0, \dots, g_s = 0$ is the *same* as the original solution set.

In the situation where $f_1, \dots, f_m$ are *linear* polynomials, a solution to the system of equations can be obtained by successively eliminating the variables $x_1, x_2, \dots$ by elementary means—using linear combinations of the original equations to eliminate the variable x_1 , then using these equations to eliminate x_2 , etc., producing a system of equations that can be easily solved (this is “Gauss-Jordan elimination” in linear algebra, cf. the exercises in Section 11.2).

The situation for polynomial equations that are nonlinear is naturally more complicated, but the basic principle is the same. If there is a nonzero polynomial in the

ideal I involving only one of the variables, say $p(x_n)$, then the last coordinate a_n is a solution of $p(x_n) = 0$. If now there is a polynomial in I involving only x_{n-1} and x_n , say $q(x_{n-1}, x_n)$, then the coordinate a_{n-1} would be a solution of $q(x_{n-1}, a_n) = 0$, etc. If we can successively find polynomials in I that eliminate the variables $x_1, x_2, \dots$ then we will be able to determine all the solutions $(a_1, \dots, a_n)$ to our original system of equations explicitly.

Finding equations that follow from the system of equations in S , i.e., finding elements of the ideal I that do not involve some of the variables, is referred to as *elimination theory*. The polynomials in I that do not involve the variables $x_1, \dots, x_i$, i.e., $I \cap F[x_{i+1}, \dots, x_n]$, is easily seen to be an ideal in $F[x_{i+1}, \dots, x_n]$ and is given a name.

Definition. If I is an ideal in $F[x_1, \dots, x_n]$ then $I_i = I \cap F[x_{i+1}, \dots, x_n]$ is called the i^{th} *elimination ideal* of I with respect to the ordering $x_1 > \dots > x_n$.

The success of using elimination to solve a system of equations depends on being able to determine the elimination ideals (and, ultimately, on whether these elimination ideals are nonzero).

The following fundamental proposition shows that if the lexicographic monomial ordering $x_1 > \dots > x_n$ is used to compute a Gröbner basis for I then the elements in the resulting basis not involving the variables $x_1, \dots, x_i$ not only determine the i^{th} elimination ideal, but in fact give a Gröbner basis for the i^{th} elimination ideal of I .

Proposition 29. (Elimination) Suppose $G = \{g_1, \dots, g_m\}$ is a Gröbner basis for the nonzero ideal I in $F[x_1, \dots, x_n]$ with respect to the lexicographic monomial ordering $x_1 > \dots > x_n$. Then $G \cap F[x_{i+1}, \dots, x_n]$ is a Gröbner basis of the i^{th} elimination ideal $I_i = I \cap F[x_{i+1}, \dots, x_n]$ of I . In particular, $I \cap F[x_{i+1}, \dots, x_n] = 0$ if and only if $G \cap F[x_{i+1}, \dots, x_n] = \emptyset$.

Proof: Denote $G_i = G \cap F[x_{i+1}, \dots, x_n]$. Then $G_i \subseteq I_i$, so by Proposition 24, to see that G_i is a Gröbner basis of I_i it suffices to see that $LT(G_i)$, the leading terms of the elements in G_i , generate $LT(I_i)$ as an ideal in $F[x_{i+1}, \dots, x_n]$. Certainly $(LT(G_i)) \subseteq LT(I_i)$ as ideals in $F[x_{i+1}, \dots, x_n]$. To show the reverse containment, let f be any element in I_i . Then $f \in I$ and since G is a Gröbner basis for I we have

$$LT(f) = a_1(x_1, \dots, x_n)LT(g_1) + \dots + a_m(x_1, \dots, x_n)LT(g_m)$$

for some polynomials $a_1, \dots, a_m \in F[x_1, \dots, x_n]$. Writing each polynomial a_i as a sum of monomial terms we see that $LT(f)$ is a sum of monomial terms of the form $ax_1^{s_1} \dots x_n^{s_n} LT(g_i)$. Since $LT(f)$ involves only the variables $x_{i+1}, \dots, x_n$, the sum of all such terms containing any of the variables $x_1, \dots, x_i$ must be 0, so $LT(f)$ is also the sum of those monomial terms only involving $x_{i+1}, \dots, x_n$. It follows that $LT(f)$ can be written as a $F[x_{i+1}, \dots, x_n]$ -linear combination of some monomial terms $LT(g_t)$ where $LT(g_t)$ does not involve the variables $x_1, \dots, x_i$. But by the choice of the ordering, if $LT(g_t)$ does not involve $x_1, \dots, x_i$, then neither do any of the other terms in g_t , i.e., $g_t \in G_i$. Hence $LT(f)$ can be written as a $F[x_{i+1}, \dots, x_n]$ -linear combination of elements $LT(G_i)$, completing the proof.

Note also that Gröbner bases can be used to eliminate any variables simply by using an appropriate monomial ordering.

Examples

- (1) The ellipse $2x^2 + 2xy + y^2 - 2x - 2y = 0$ intersects the circle $x^2 + y^2 = 1$ in two points. To find them we compute a Gröbner basis for the ideal $I = (2x^2 + 2xy + y^2 - 2x - 2y, x^2 + y^2 - 1) \subset \mathbb{R}[x, y]$ using the lexicographic monomial order $x > y$ to eliminate x , obtaining $g_1 = 2x + y^2 + 5y^3 - 2$ and $g_2 = 5y^4 - 4y^3$. Hence $5y^4 = 4y^3$ and $y = 0$ or $y = 4/5$. Substituting these values into $g_1 = 0$ and solving for x we find the two intersection points are $(1, 0)$ and $(-3/5, 4/5)$.

Instead using the lexicographic monomial order $y > x$ to eliminate y results in the Gröbner basis $\{y^2 + x^2 - 1, 2yx - 2y + x^2 - 2x + 1, 5x^3 - 7x^2 - x + 3\}$. Then $5x^3 - 7x^2 - x + 3 = (x - 1)^2(5x + 3)$ shows that x is 1 or $-3/5$ and we obtain the same solutions as before, although with more effort.

- (2) In the previous example the solutions could also have been found by elementary means. Consider now the solutions in $\mathbb{C}$ to the system of two equations

$$x^3 - 2xy + y^3 = 0 \quad \text{and} \quad x^5 - 2x^2y^2 + y^5 = 0.$$

Computing a Gröbner basis for the ideal generated by $f_1 = x^3 - 2xy + y^3$ and $f_2 = x^5 - 2x^2y^2 + y^5$ with respect to the lexicographic monomial order $x > y$ we obtain the basis

$$g_1 = x^3 - 2xy + y^3$$

$$g_2 = 200xy^2 + 193y^9 + 158y^8 - 45y^7 - 456y^6 + 50y^5 - 100y^4$$

$$g_3 = y^{10} - y^8 - 2y^7 + 2y^6.$$

Any solution to our original equations would satisfy $g_1 = g_2 = g_3 = 0$. Since $g_3 = y^6(y - 1)^2(y^2 + 2y + 2)$, we have $y = 0$, $y = 1$ or $y = -1 \pm i$. Since $g_1(x, 0) = x^3$ and $g_2(x, 0) = 0$, we see that $(0, 0)$ is the only solution with $y = 0$. Since $g_1(x, 1) = x^3 - 2x + 1$ and $g_2(x, 1) = 200(x - 1)$ have only $x = 1$ as a common zero, the only solution with $y = 1$ is $(1, 1)$. Finally,

$$g_1(x, -1 \pm i) = x^3 + (2 \mp 2i)x + (2 \pm 2i)$$

$$g_2(x, -1 \pm i) = -400i(x + 1 \pm i),$$

and a quick check shows the common zero $x = -1 \mp i$ when $y = -1 \pm i$, respectively. Hence, there are precisely four solutions to the original pair of equations, namely

$$(x, y) = (0, 0), \quad (1, 1), \quad (-1 + i, -1 - i), \quad \text{or} \quad (-1 - i, -1 + i).$$

- (3) Consider the solutions in $\mathbb{C}$ to the system of equations

$$x + y + z = 1$$

$$x^2 + y^2 + z^2 = 2$$

$$x^3 + y^3 + z^3 = 3.$$

The reduced Gröbner basis with respect to the lexicographic ordering $x > y > z$ is

$$\{x + y + z - 1, \quad y^2 + yz - y + z^2 - z - (1/2), \quad z^3 - z^2 - (1/2)z - (1/6)\}$$

and so z is a root of the polynomial $t^3 - t^2 - (1/2)t - (1/6)$ (by symmetry, also x and y are roots of this same polynomial). For each of the three roots of this polynomial, there are two values of y and one corresponding value of x making the first two polynomials in the Gröbner basis equal to 0. The resulting six solutions are quickly checked to be the three distinct roots of the polynomial $t^3 - t^2 - (1/2)t - (1/6)$ (which is irreducible over $\mathbb{Q}$) in some order.

As the previous examples show, the study of solutions to systems of polynomial equations $f_1 = 0, f_2 = 0, \dots, f_m = 0$ is intimately related to the study of the ideal $I = (f_1, f_2, \dots, f_m)$ the polynomials generate in $F[x_1, \dots, x_n]$. This fundamental connection is the starting point for the important and active branch of mathematics called “algebraic geometry”, introduced in Chapter 15, where additional applications of Gröbner bases are given.

We close this section by showing how to compute the basic set-theoretic operations of sums, products and intersections of ideals in polynomial rings. Suppose $I = (f_1, \dots, f_s)$ and $J = (h_1, \dots, h_t)$ are two ideals in $F[x_1, \dots, x_n]$. Then $I + J = (f_1, \dots, f_s, h_1, \dots, h_t)$ and $IJ = (f_1 h_1, \dots, f_i h_j, \dots, f_s h_t)$. The following proposition shows how to compute the intersection of any two ideals.

Proposition 30. If I and J are any two ideals in $F[x_1, \dots, x_n]$ then $tI + (1 - t)J$ is an ideal in $F[t, x_1, \dots, x_n]$ and $I \cap J = (tI + (1 - t)J) \cap F[x_1, \dots, x_n]$. In particular, $I \cap J$ is the first elimination ideal of $tI + (1 - t)J$ with respect to the ordering $t > x_1 > \dots > x_n$.

Proof: First, tI and $(1 - t)J$ are clearly ideals in $F[x_1, \dots, x_n, t]$, so also their sum $tI + (1 - t)J$ is an ideal in $F[x_1, \dots, x_n, t]$. If $f \in I \cap J$, then $f = tf + (1 - t)f$ shows $I \cap J \subseteq (tI + (1 - t)J) \cap F[x_1, \dots, x_n]$. Conversely, suppose $f = tf_1 + (1 - t)f_2$ is an element of $F[x_1, \dots, x_n]$, where $f_1 \in I$ and $f_2 \in J$. Then $t(f_1 - f_2) = f - f_2 \in F[x_1, \dots, x_n]$ shows that $f_1 - f_2 = 0$ and $f = f_2$, so $f = f_1 = f_2 \in I \cap J$. Since $I \cap J = (tI + (1 - t)J) \cap F[x_1, \dots, x_n]$, $I \cap J$ is the first elimination ideal of $tI + (1 - t)J$ with respect to the ordering $t > x_1 > \dots > x_n$.

We have $tI + (1 - t)J = (tf_1, \dots, tf_s, (1 - t)h_1, \dots, (1 - t)h_t)$ if $I = (f_1, \dots, f_s)$ and $J = (h_1, \dots, h_t)$. By Proposition 29, the elements not involving t in a Gröbner basis for this ideal in $F[t, x_1, \dots, x_n]$, computed for the lexicographic monomial ordering $t > x_1 > \dots > x_n$, give a Gröbner basis for the ideal $I \cap J$ in $F[x_1, \dots, x_n]$.

Example

Let $I = (x, y)^2 = (x^2, xy, y^2)$ and let $J = (x)$. For the lexicographic monomial ordering $t > x > y$ the reduced Gröbner basis for $tI + (1 - t)J$ in $F[t, x, y]$ is $\{tx - x, ty^2, x^2, xy\}$ and so $I \cap J = (x^2, xy)$.

EXERCISES

1. Suppose I is an ideal in $F[x_1, \dots, x_n]$ generated by a (possibly infinite) set $\mathcal{S}$ of polynomials. Prove that a finite subset of the polynomials in $\mathcal{S}$ suffice to generate I . [Use Theorem 21 to write $I = (f_1, \dots, f_m)$ and then write each $f_i \in I$ using polynomials in $\mathcal{S}$.]
2. Let $\geq$ be any monomial ordering.
 - (a) Prove that $LT(fg) = LT(f)LT(g)$ and $\partial(fg) = \partial(f) + \partial(g)$ for any nonzero polynomials f and g .
 - (b) Prove that $\partial(f + g) \leq \max(\partial(f), \partial(g))$ with equality if $\partial(f) \neq \partial(g)$.

- (c) Prove that $m \geq 1$ for every monomial m .
- (d) Prove that if m_1 divides m_2 then $m_2 \geq m_1$. Deduce that the leading term of a polynomial does not divide any of its lower order terms.
3. Prove that if $\geq$ is any total or partial ordering on a nonempty set then the following are equivalent:
- Every nonempty subset contains a minimum element.
 - There is no infinite strictly decreasing sequence $a_1 > a_2 > a_3 > \dots$ (this is called the *descending chain condition* or *D.C.C.*).
- Deduce that General Polynomial Division always terminates in finitely many steps.
4. Let $\geq$ be a monomial ordering, and for monomials m_1, m_2 define $m_1 \geq_g m_2$ if either $\deg m_1 > \deg m_2$, or $\deg m_1 = \deg m_2$ and $m_1 \geq m_2$.
- Prove that $\geq_g$ is also a monomial ordering. (The relation $\geq_g$ is called the *grading* of $\geq$. An ordering in which the most important criterion for comparison is degree is sometimes called a *graded* or a *degree* ordering, so this exercise gives a method for constructing graded orderings.)
 - The grading of the lexicographic ordering $x_1 > \dots > x_n$ is called the *grlex* monomial ordering. Show that $x_2^4 > x_1^2 x_2^2 > x_1 x_2^3 > x_2^3 > x_1$ with respect to the grlex ordering and $x_1^2 x_2 > x_1 x_2^2 > x_1 > x_2^4 > x_2^3$ with respect to the lexicographic ordering.
5. The *grevlex* monomial ordering is defined by first choosing an ordering of the variables $\{x_1, x_2, \dots, x_n\}$, then defining $m_1 \geq m_2$ for monomials m_1, m_2 if either $\deg m_1 > \deg m_2$ or $\deg m_1 = \deg m_2$ and the first exponent of $x_n, x_{n-1}, \dots, x_1$ (in that order) where m_1 and m_2 differ is *smaller* in m_1 .
- Prove that grevlex is a monomial ordering that satisfies $x_1 > x_2 > \dots > x_n$.
 - Prove that the grevlex ordering on $F[x_1, x_2]$ with respect to $\{x_1, x_2\}$ is the graded lexicographic ordering with $x_1 > x_2$, but that the grevlex ordering on $F[x_1, x_2, x_3]$ is not the grading of any lexicographic ordering.
 - Show that $x_1 x_2^2 x_3 > x_1^2 x_3^2 > x_2^2 x_3^2 > x_2 x_3^2 > x_1 x_2 > x_2^2 > x_1 x_3 > x_3^2 > x_1 > x_2$ for the grevlex monomial ordering with respect to $\{x_1, x_2, x_3\}$.
6. Show that $x^3 y > x^3 z^2 > x^3 z > x^2 y^2 z > x^2 y > xz^2 > y^2 z^2 > y^2 z$ with respect to the lexicographic monomial ordering $x > y > z$. Show that for the corresponding grlex monomial ordering $x^3 z^2 > x^2 y^2 z > x^3 y > x^3 z > y^2 z^2 > x^2 y > xz^2 > y^2 z$, and that $x^2 y^2 z > x^3 z^2 > x^3 y > x^3 z > y^2 z^2 > x^2 y > y^2 z > xz^2$ for the grevlex monomial ordering with respect to $\{x, y, z\}$.
7. Order the monomials $x^2 z, x^2 y^2 z, xy^2 z, x^3 y, x^3 z^2, x^2, x^2 yz^2, x^2 z^2$ for the lexicographic monomial ordering $x > y > z$, for the corresponding grlex monomial order, and for the grevlex monomial ordering with respect to $\{x, y, z\}$.
8. Show there are $n!$ distinct lexicographic monomial orderings on $F[x_1, \dots, x_n]$. Show similarly that there are $n!$ distinct grlex and grevlex monomial orderings.
9. It can be shown that any monomial ordering on $F[x_1, \dots, x_n]$ may be obtained as follows. For $k \leq n$ let $v_1, v_2, \dots, v_k$ be nonzero vectors in Euclidean n -space, $\mathbb{R}^n$, that are pairwise orthogonal: $v_i \cdot v_j = 0$ for all $i \neq j$, where $\cdot$ is the usual dot product, and suppose also that all the coordinates of v_i are nonnegative. Define an order, $\geq$, on monomials by $m_1 > m_2$ if and only if for some $t \leq k$ we have $v_i \cdot \partial(m_1) = v_i \cdot \partial(m_2)$ for all $i \in \{1, 2, \dots, t-1\}$ and $v_t \cdot \partial(m_1) > v_t \cdot \partial(m_2)$.
- Let $k = n$ and let $v_i = (0, \dots, 0, 1, 0, \dots, 0)$ with 1 in the i^{th} position. Show that $\geq$ defines the lexicographic order with $x_1 > x_2 > \dots > x_n$.
 - Let $k = n$ and define $v_1 = (1, 1, \dots, 1)$ and $v_i = (1, 1, \dots, 1, -n+i-1, 0, \dots, 0)$,

where there are $i - 2$ trailing zeros, $2 \leq i \leq n$. Show that $\geq$ defines the grlex order with respect to $\{x_1, \dots, x_n\}$.

10. Suppose I is a monomial ideal generated by monomials $m_1, \dots, m_k$. Prove that the polynomial $f \in F[x_1, \dots, x_n]$ is in I if and only if every monomial term f_i of f is a multiple of one of the m_j . [For polynomials $a_1, \dots, a_k \in F[x_1, \dots, x_n]$ expand the polynomial $a_1 m_1 + \dots + a_k m_k$ and note that every monomial term is a multiple of at least one of the m_j .] Show that $x^2 y z + 3 x y^2$ is an element of the ideal $I = (x y z, y^2) \subset F[x, y, z]$ but is not an element of the ideal $I' = (x z^2, y^2)$.
11. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$ and suppose $\{g_1, \dots, g_m\}$ is a Gröbner basis for the ideal I in R . Prove that $h \in LT(I)$ if and only if h is a sum of monomial terms each divisible by some $LT(g_i)$, $1 \leq i \leq m$. [Use the previous exercise.]
12. Suppose I is a monomial ideal with monomial generators $g_1, \dots, g_m$. Use the previous exercise to prove directly that $\{g_1, \dots, g_m\}$ is a Gröbner basis for I .
13. Suppose I is a monomial ideal with monomial generators $g_1, \dots, g_m$. Use Buchberger's Criterion to prove that $\{g_1, \dots, g_m\}$ is a Gröbner basis for I .
14. Suppose I is a monomial ideal in $R = F[x_1, \dots, x_n]$ and suppose $\{m_1, \dots, m_k\}$ is a minimal set of monomials generating I , i.e., each m_i is a monomial and no proper subset of $\{m_1, \dots, m_k\}$ generates I . Prove that the m_i , $1 \leq i \leq k$ are unique. [Use Exercise 10.]
15. Fix a monomial ordering on $R = F[x_1, \dots, x_n]$.
 - (a) Prove that $\{g_1, \dots, g_m\}$ is a minimal Gröbner basis for the ideal I in R if and only if $\{LT(g_1), \dots, LT(g_m)\}$ is a minimal generating set for $LT(I)$.
 - (b) Prove that the leading terms of a minimal Gröbner basis for I are uniquely determined and the number of elements in any two minimal Gröbner bases for I is the same. [Use (a) and the previous exercise.]
16. Fix a monomial ordering on $F[x_1, \dots, x_n]$ and suppose $G = \{g_1, \dots, g_m\}$ is a set of generators for the nonzero ideal I . Show that if $S(g_i, g_j) \not\equiv 0 \pmod{G}$ then the ideal $(LT(g_1), \dots, LT(g_m), LT(S(g_i, g_j)))$ is strictly larger than the ideal $(LT(g_1), \dots, LT(g_m))$. Conclude that the algorithm for computing a Gröbner basis described following Proposition 26 terminates after a finite number of steps. [Use Exercise 1.]
17. Fix the lexicographic ordering $x > y$ on $F[x, y]$. Use Buchberger's Criterion to show that $\{x^2 y - y^2, x^3 - x y\}$ is a Gröbner basis for the ideal $I = (x^2 y - y^2, x^3 - x y)$.
18. Show $\{x - y^3, y^5 - y^6\}$ is the reduced Gröbner basis for the ideal $I = (x - y^3, -x^2 + x y^2)$ with respect to the lexicographic ordering defined by $x > y$ in $F[x, y]$.
19. Fix the lexicographic ordering $x > y$ on $F[x, y]$.
 - (a) Show that $\{x^3 - y, x^2 y - y^2, x y^2 - y^2, y^3 - y^2\}$ is the reduced Gröbner basis for the ideal $I = (-x^3 + y, x^2 y - y^2)$.
 - (b) Determine whether the polynomial $f = x^6 - x^5 y$ is an element of the ideal I .
20. Fix the lexicographic ordering $x > y > z$ on $F[x, y, z]$. Show that $\{x^2 + x y + z, x y z + z^2, x z^2, z^3\}$ is the reduced Gröbner basis for the ideal $I = (x^2 + x y + z, x y z + z^2)$ and in particular conclude that the leading term ideal $LT(I)$ requires four generators.
21. Fix the lexicographic ordering $x > y$ on $F[x, y]$. Use Buchberger's Criterion to show that $\{x^2 y - y^2, x^3 - x y\}$ is a Gröbner basis for the ideal $I = (x^2 y - y^2, x^3 - x y)$.
22. Let $I = (x^2 - y, x^2 y - z)$ in $F[x, y, z]$.
 - (a) Show that $\{x^2 - y, y^2 - z\}$ is the reduced Gröbner basis for I with respect to the lexicographic ordering defined by $x > y > z$.
 - (b) Show that $\{x^2 - y, z - y^2\}$ is the reduced Gröbner basis for I with respect to the

lexicographic ordering defined by $z > x > y$ (note these are essentially the same polynomials as in (a)).

(c) Show that $\{y - x^2, z - x^4\}$ is the reduced Gröbner basis for I with respect to the lexicographic ordering defined by $z > y > x$.

23. Show that the ideals $I = (x^2y + xy^2 - 2y, x^2 + xy - x + y^2 - 2y, xy^2 - x - y + y^3)$ and $J = (x - y^2, xy - y, x^2 - y)$ in $F[x, y]$ are equal.
24. Use reduced Gröbner bases to show that the ideal $I = (x^3 - yz, yz + y)$ and the ideal $J = (x^3z + x^3, x^3 + y)$ in $F[x, y, z]$ are equal.
25. Show that the reduced Gröbner basis using the lexicographic ordering $x > y$ for the ideal $I = (x^2 + xy^2, x^2 - y^3, y^3 - y^2)$ is $\{x^2 - y^2, y^3 - y^2, xy^2 + y^2\}$.
26. Show that the reduced Gröbner basis for the ideal $I = (xy + y^2, x^2y + xy^2 + x^2)$ is $\{x^2, xy + y^2, y^3\}$ with respect to the lexicographic ordering $x > y$ and is $\{y^2 + yx, x^2\}$ with respect to the lexicographic ordering $y > x$.

There are generally substantial differences in computational complexity when using different monomial orders. The grevlex monomial ordering often provides the most efficient computation and produces simpler polynomials.

27. Show that $\{x^3 - y^3, x^2 + xy^2 + y^4, x^2y + xy^3 + y^2\}$ is a reduced Gröbner basis for the ideal I in the example following Corollary 28 with respect to the grlex monomial ordering. (Note that while this gives three generators for I rather than two for the lexicographic ordering as in the example, the degrees are smaller.)
28. Let $I = (x^4 - y^4 + z^3 - 1, x^3 + y^2 + z^2 - 1)$. Show that there are five elements in a reduced Gröbner basis for I with respect to the lexicographic ordering with $x > y > z$ (the maximum degree among the five generators is 12 and the maximum number of monomial terms among the five generators is 35), that there are two elements for the lexicographic ordering $y > z > x$ (maximum degree is 6 and maximum number of terms is 8), and that $\{x^3 + y^2 + z^2 - 1, xy^2 + xz^2 - x + y^4 - z^3 + 1\}$ is the reduced Gröbner basis for the grevlex monomial ordering.
29. Solve the system of equations $x^2 - yz = 3, y^2 - xz = 4, z^2 - xy = 5$ over $\mathbb{C}$.
30. Find a Gröbner basis for the ideal $I = (x^2 + xy + y^2 - 1, x^2 + 4y^2 - 4)$ for the lexicographic ordering $x > y$ and use it to find the four points of intersection of the ellipse $x^2 + xy + y^2 = 1$ with the ellipse $x^2 + 4y^2 = 4$ in $\mathbb{R}^2$.
31. Use Gröbner bases to find all six solutions to the system of equations $2x^3 + 2x^2y^2 + 3y^3 = 0$ and $3x^5 + 2x^3y^3 + 2y^5 = 0$ over $\mathbb{C}$.
32. Use Gröbner bases to show that $(x, z) \cap (y^2, x - yz) = (xy, x - yz)$ in $F[x, y, z]$.
33. Use Gröbner bases to compute the intersection of the ideals $(x^3y - xy^2 + 1, x^2y^2 - y^3 - 1)$ and $(x^2 - y^2, x^3 + y^3)$ in $F[x, y]$.

The following four exercises deal with the *ideal quotient* of two ideals I and J in a ring R .

Definition. The *ideal quotient* $(I : J)$ of two ideals I, J in a ring R is the ideal

$$(I : J) = \{r \in R \mid rJ \subseteq I\}.$$

34. (a) Suppose R is an integral domain, $0 \neq f \in R$ and I is an ideal in R . Show that if $\{g_1, \dots, g_s\}$ are generators for the ideal $I \cap (f)$, then $\{g_1/f, \dots, g_s/f\}$ are generators for the ideal quotient $(I : (f))$.
- (b) If I is an ideal in the commutative ring R and $f_1, \dots, f_s \in R$, show that the ideal quotient $(I : (f_1, \dots, f_s))$ is the ideal $\bigcap_{i=1}^s (I : (f_i))$.

35. If $I = (x^2y + z^3, x + y^3 - z, 2y^4z - yz^2 - z^3)$ and $J = (x^2y^5, x^3z^4, y^3z^7)$ in $\mathbb{Q}[x, y, z]$ show $(I : J)$ is the ideal $(z^2, y + z, x - z)$. [Use the previous exercise and Proposition 30.]
36. Suppose that K is an ideal in R , that I is an ideal containing K , and J is any ideal. If $\bar{I}$ and $\bar{J}$ denote the images of I and J in the quotient ring R/K , show that $(\bar{I} : \bar{J}) = (\bar{I} : \bar{J})$ where $(\bar{I} : \bar{J})$ is the image in R/K of the ideal quotient $(I : J)$.
37. Let K be the ideal $(y^5 - z^4)$ in $R = \mathbb{Q}[y, z]$. For each of the following pairs of ideals I and J , use the previous two exercises together with Proposition 30 to verify the ideal quotients $(\bar{I} : \bar{J})$ in the ring R/K :
- $I = (y^3, y^5 - z^4), J = (z), (\bar{I} : \bar{J}) = (\bar{y}^3, \bar{z}^3)$.
 - $I = (y^3, z, y^5 - z^4), J = (y), (\bar{I} : \bar{J}) = (\bar{y}^2, \bar{z})$.
 - $I = (y, y^3, z, y^5 - z^4), J = (1), (\bar{I} : \bar{J}) = (\bar{y}, \bar{z})$.

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \dots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, M and N are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets I and J respectively.

38. Prove that the sum and product of two monomial ideals is a monomial ideal by showing that $M + N = (m_i, n_j \mid i \in I, j \in J)$, and $MN = (m_i n_j \mid i \in I, j \in J)$.
39. Show that if $\{M_s \mid s \in S\}$ is any nonempty collection of monomial ideals that is totally ordered under inclusion then $\bigcup_{s \in S} M_s$ is a monomial ideal. (In particular, the union of any increasing sequence of monomial ideals is a monomial ideal, cf. Exercise 19, Section 7.3.)
40. Prove that the intersection of two monomial ideals is a monomial ideal by showing that $M \cap N = (e_{i,j} \mid i \in I, j \in J)$, where $e_{i,j}$ is the least common multiple of m_i and n_j . [Use Exercise 10.]
41. Prove that for any monomial n , the ideal quotient $(M : (n))$ is $(m_i/d_i \mid i \in I)$, where d_i is the greatest common divisor of m_i and n (cf. Exercise 34). Show that if N is finitely generated, then the ideal quotient $(M : N)$ of two monomial ideals is a monomial ideal.
42. (a) Show that M is a monomial prime ideal if and only if $M = (S)$ for some subset of S of $\{x_1, x_2, \dots, x_n\}$. (In particular, there are only finitely many monomial prime ideals, and each is finitely generated.)
(b) Show that $(x_1, \dots, x_n)$ is the only monomial maximal ideal.
43. (Dickson's Lemma—a special case of Hilbert's Basis Theorem) Prove that every monomial ideal in $F[x_1, \dots, x_n]$ is finitely generated as follows.
Let $\mathcal{S} = \{N \mid N \text{ is a monomial ideal that is not finitely generated}\}$, and assume by way of contradiction $\mathcal{S} \neq \emptyset$.
(a) Show that $\mathcal{S}$ contains a maximal element M . [Use Exercise 30 and Zorn's Lemma.]
(b) Show that there are monomials x, y not in M with $xy \in M$. [Use Exercise 33(a).]
(c) For x as in (b), show that M contains a finitely generated monomial ideal M_0 such that $M_0 + (x) = M + (x)$ and $M = M_0 + (x)(M : (x))$, where $(M : (x))$ is the (monomial) ideal defined in Exercise 32, and $(x)(M : (x))$ is the product of these two ideals. Deduce that M is finitely generated, a contradiction which proves $\mathcal{S} = \emptyset$. [Use the maximality of M and previous exercises.]
44. If I is a nonzero ideal in $F[x_1, \dots, x_n]$, use Dickson's Lemma to prove that $LT(I)$ is finitely generated. Conclude that I has a Gröbner basis and deduce Hilbert's Basis Theorem. [cf. Proposition 24.]

45. (*n-colorings of graphs*) A finite graph $\mathcal{G}$ of size N is a set of vertices $i \in \{1, 2, \dots, N\}$ and a collection of edges (i, j) connecting vertex i with vertex j . An n -coloring of $\mathcal{G}$ is an assignment of one of n colors to each vertex in such a way that vertices connected by an edge have distinct colors. Let F be any field containing at least n elements. If we introduce a variable x_i for each vertex i and represent the n colors by choosing a set S of n distinct elements from F , then an n -coloring of $\mathcal{G}$ is equivalent to assigning a value $x_i = \alpha_i$ for each $i = 1, 2, \dots, N$ where $\alpha_i \in S$ and $\alpha_i \neq \alpha_j$ if (i, j) is an edge in $\mathcal{G}$. If $f(x) = \prod_{\alpha \in S} (x - \alpha)$ is the polynomial in $F[x]$ of degree n whose roots are the elements in S , then $x_i = \alpha_i$ for some $\alpha_i \in S$ is equivalent to the statement that x_i is a solution to the equation $f(x_i) = 0$. The statement $\alpha_i \neq \alpha_j$ is then the statement that $f(x_i) = f(x_j)$ but $x_i \neq x_j$, so x_i and x_j satisfy the equation $g(x_i, x_j) = 0$, where $g(x_i, x_j)$ is the polynomial $(f(x_i) - f(x_j))/(x_i - x_j)$ in $F[x_i, x_j]$. It follows that finding an n -coloring of $\mathcal{G}$ is equivalent to solving the system of equations

$$\begin{cases} f(x_i) = 0, & \text{for } i = 1, 2, \dots, N. \\ g(x_i, x_j) = 0, & \text{for all edges } (i, j) \text{ in } \mathcal{G} \end{cases}$$

(note also we may use any polynomial g satisfying $\alpha_i \neq \alpha_j$ if $g(\alpha_i, \alpha_j) = 0$). It follows by “Hilbert’s Nullstellensatz” (cf. Corollary 33 in Section 15.3) that this system of equations has a solution, hence $\mathcal{G}$ has an n -coloring, unless the ideal I in $F[x_1, x_2, \dots, x_N]$ generated by the polynomials $f(x_i)$ for $i = 1, 2, \dots, N$, together with the polynomials $g(x_i, x_j)$ for all the edges (i, j) in the graph $\mathcal{G}$, is not a proper ideal. This in turn is equivalent to the statement that the reduced Gröbner basis for I (with respect to any monomial ordering) is simply $\{1\}$. Further, when an n -coloring does exist, solving this system of equations as in the examples following Proposition 29 provides an explicit coloring for $\mathcal{G}$.

There are many possible choices of field F and set S . For example, use any field F containing a set S of distinct n^{th} roots of unity, in which case $f(x) = x^n - 1$ and we may take $g(x_i, x_j) = (x_i^n - x_j^n)/(x_i - x_j) = x_i^{n-1} + x_i^{n-2}x_j + \dots + x_ix_j^{n-2} + x_j^{n-1}$, or use any subset S of $F = \mathbb{F}_p$ with a prime $p \geq n$ (in the special case $n = p$, then, by Fermat’s Little Theorem, we have $f(x) = x^p - x$ and $g(x_i, x_j) = (x_i^p - x_j^p)/(x_i - x_j) = x_i^{p-1} + x_i^{p-2}x_j + \dots + x_ix_j^{p-2} + x_j^{p-1}$).

- (a) Consider a possible 3-coloring of the graph $\mathcal{G}$ with eight vertices and 14 edges $(1, 3), (1, 4), (1, 5), (2, 4), (2, 7), (2, 8), (3, 4), (3, 6), (3, 8), (4, 5), (5, 6), (6, 7), (6, 8), (7, 8)$. Take $F = \mathbb{F}_3$ with ‘colors’ $0, 1, 2 \in \mathbb{F}_3$ and suppose vertex 1 is colored by 0. In this case $f(x) = x(x-1)(x-2) = x^3 - x \in \mathbb{F}_3[x]$ and $g(x_i, x_j) = x_i^2 + x_ix_j + x_j^2 - 1$. If I is the ideal generated by $x_1, x_i^3 - x_i, 2 \leq i \leq 8$ and $g(x_i, x_j)$ for the edges (i, j) in $\mathcal{G}$, show that the reduced Gröbner basis for I with respect to the lexicographic monomial ordering $x_1 > x_2 > \dots > x_8$ is $\{x_1, x_2, x_3 + x_8, x_4 + 2x_8, x_5 + x_8, x_6, x_7 + x_8, x_8^2 + 2\}$. Deduce that $\mathcal{G}$ has two distinct 3-colorings, determined by the coloring of vertex 8 (which must be colored by a nonzero element in $\mathbb{F}_3$), and exhibit the colorings of $\mathcal{G}$.
- Show that if the edge $(3, 7)$ is added to $\mathcal{G}$ then the graph cannot be 3-colored.
- (b) Take $F = \mathbb{F}_5$ with four ‘colors’ $1, 2, 3, 4 \in \mathbb{F}_5$, so $f(x) = x^4 - 1$ and we may use $g(x_i, x_j) = x_i^3 + x_i^2x_j + x_ix_j^2 + x_j^3$. Show that the graph $\mathcal{G}$ with five vertices having 9 edges $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)$ (the “complete graph on five vertices” with one edge removed) can be 4-colored but cannot be 3-colored.
- (c) Use Gröbner bases to show that the graph $\mathcal{G}$ with nine vertices and 22 edges $(1, 4), (1, 6), (1, 7), (1, 8), (2, 3), (2, 4), (2, 6), (2, 7), (3, 5), (3, 7), (3, 9), (4, 5), (4, 6), (4, 7), (4, 9), (5, 6), (5, 7), (5, 8), (5, 9), (6, 7), (6, 9), (7, 8)$ has precisely four 4-colorings up to a permutation of the colors (so a total of 96 total 4-colorings). Show that if the edge $(1, 5)$ is added then $\mathcal{G}$ cannot be 4-colored.

Part III

MODULES AND VECTOR SPACES

In Part III we study the mathematical objects called modules. The use of modules was pioneered by one of the most prominent mathematicians of the first part of this century, Emmy Noether, who led the way in demonstrating the power and elegance of this structure. We shall see that vector spaces are just special types of modules which arise when the underlying ring is a field. If R is a ring, the definition of an R -module M is closely analogous to the definition of a group action where R plays the role of the group and M the role of the set. The additional axioms for a module require that M itself have more structure (namely that M be an abelian group). Modules are the “representation objects” for rings, i.e., they are, by definition, algebraic objects on which rings act. As the theory develops it will become apparent how the structure of the ring R (in particular, the structure and wealth of its ideals) is reflected by the structure of its modules and vice versa in the same way that the structure of the collection of normal subgroups of a group was reflected by its permutation representations.

Introduction to Module Theory

10.1 BASIC DEFINITIONS AND EXAMPLES

We start with the definition of a module.

Definition. Let R be a ring (not necessarily commutative nor with 1). A *left R -module* or a *left module over R* is a set M together with

- (1) a binary operation $+$ on M under which M is an abelian group, and
- (2) an action of R on M (that is, a map $R \times M \rightarrow M$) denoted by rm , for all $r \in R$ and for all $m \in M$ which satisfies
 - (a) $(r + s)m = rm + sm$, for all $r, s \in R, m \in M$,
 - (b) $(rs)m = r(sm)$, for all $r, s \in R, m \in M$, and
 - (c) $r(m + n) = rm + rn$, for all $r \in R, m, n \in M$.

If the ring R has a 1 we impose the additional axiom:

- (d) $1m = m$, for all $m \in M$.

The descriptor “left” in the above definition indicates that the ring elements appear on the left; “right” R -modules can be defined analogously. If the ring R is *commutative* and M is a left R -module we can make M into a right R -module by defining $mr = rm$ for $m \in M$ and $r \in R$. If R is not commutative, axiom 2(b) in general will not hold with this definition (so not every left R -module is also a right R -module). Unless explicitly mentioned otherwise the term “module” will always mean “left module.” Modules satisfying axiom 2(d) are called *unital* modules and in this book all our modules will be unital (this is to avoid “pathologies” such as having $rm = 0$ for all $r \in R$ and $m \in M$).

When R is a field F the axioms for an R -module are precisely the same as those for a vector space over F , so that

modules over a field F and vector spaces over F are the same.

Before giving other examples of R -modules we record the obvious definition of submodules.

Definition. Let R be a ring and let M be an R -module. An *R -submodule* of M is a subgroup N of M which is closed under the action of ring elements, i.e., $rn \in N$, for all $r \in R, n \in N$.

Submodules of M are therefore just subsets of M which are themselves modules under the restricted operations. In particular, if $R = F$ is a field, submodules are the same as subspaces. Every R -module M has the two submodules M and 0 (the latter is called the *trivial submodule*).

Examples

- (1) Let R be any ring. Then $M = R$ is a left R -module, where the action of a ring element on a module element is just the usual multiplication in the ring R (similarly, R is a right module over itself). In particular, every field can be considered as a (1-dimensional) vector space over itself. When R is considered as a left module over itself in this fashion, the submodules of R are precisely the left ideals of R (and if R is considered as a right R -module over itself, its submodules are the right ideals). Thus if R is not commutative it has a left and right module structure over itself and these structures may be different (e.g., the submodules may be different) — Exercise 21 at the end of this section gives a specific example of this.
- (2) Let $R = F$ be a field. As noted above, every vector space over F is an F -module and vice versa. Let $n \in \mathbb{Z}^+$ and let

$$F^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in F, \text{ for all } i\}$$

(called *affine n -space over F*). Make F^n into a vector space by defining addition and scalar multiplication componentwise:

$$\begin{aligned}(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) &= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \\ \alpha(a_1, \dots, a_n) &= (\alpha a_1, \dots, \alpha a_n), \quad \alpha \in F.\end{aligned}$$

As in the case of Euclidean n -space (i.e., when $F = \mathbb{R}$), affine n -space is a vector space of dimension n over F (we shall discuss the notion of dimension more thoroughly in the next chapter).

- (3) Let R be a ring with 1 and let $n \in \mathbb{Z}^+$. Following Example 2 define

$$R^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in R, \text{ for all } i\}.$$

Make R^n into an R -module by componentwise addition and multiplication by elements of R in the same manner as when R was a field. The module R^n is called *the free module of rank n over R* . (We shall see shortly that free modules have the same “universal property” in the context of R -modules that free groups were seen to have in Section 6.3. We shall also soon discuss direct products of R -modules.) An obvious submodule of R^n is given by the i^{th} component, namely the set of n -tuples with arbitrary ring elements in the i^{th} component and zeros in the j^{th} component for all $j \neq i$.

- (4) The same abelian group may have the structure of an R -module for a number of different rings R and each of these module structures may carry useful information. Specifically, if M is an R -module and S is a subring of R with $1_S = 1_R$, then M is automatically an S -module as well. For instance the field $\mathbb{R}$ is an $\mathbb{R}$ -module, a $\mathbb{Q}$ -module and a $\mathbb{Z}$ -module.
- (5) If M is an R -module and for some (2-sided) ideal I of R , $am = 0$, for all $a \in I$ and all $m \in M$, we say M is *annihilated by I* . In this situation we can make M into an (R/I) -module by defining an action of the quotient ring R/I on M as follows: for each $m \in M$ and coset $r + I$ in R/I let

$$(r + I)m = rm.$$

Since $am = 0$ for all $a \in I$ and all $m \in M$ this is well defined and one easily checks that it makes M into an (R/I) -module. In particular, when I is a maximal ideal in the commutative ring R and $IM = 0$, then M is a vector space over the field R/I (cf. the following example).

The next example is of sufficient importance as to be singled out. It will form the basis for our proof of the Fundamental Theorem of Finitely Generated Abelian Groups in Chapter 12.

Example: ($\mathbb{Z}$ -modules)

Let $R = \mathbb{Z}$, let A be any abelian group (finite or infinite) and write the operation of A as $+$. Make A into a $\mathbb{Z}$ -module as follows: for any $n \in \mathbb{Z}$ and $a \in A$ define

$$na = \begin{cases} a + a + \cdots + a & (n \text{ times}) & \text{if } n > 0 \\ 0 & & \text{if } n = 0 \\ -a - a - \cdots - a & (-n \text{ times}) & \text{if } n < 0 \end{cases}$$

(here 0 is the identity of the additive group A). This definition of an action of the integers on A makes A into a $\mathbb{Z}$ -module, and the module axioms show that this is the only possible action of $\mathbb{Z}$ on A making it a (unital) $\mathbb{Z}$ -module. Thus every abelian group is a $\mathbb{Z}$ -module. Conversely, if M is any $\mathbb{Z}$ -module, a fortiori M is an abelian group, so

$\mathbb{Z}$ -modules are the same as abelian groups.

Furthermore, it is immediate from the definition that

$\mathbb{Z}$ -submodules are the same as subgroups.

Note that for the cyclic group $\langle a \rangle$ written multiplicatively the additive notation na becomes a^n , that is, we have all along been using the fact that $\langle a \rangle$ is a right $\mathbb{Z}$ -module (checking that this “exponential” notation satisfies the usual laws of exponents is equivalent to checking the $\mathbb{Z}$ -module axioms — this was given as an exercise at the end of Section 1.1). Note that since $\mathbb{Z}$ is commutative these definitions of left and right actions by ring elements give the same module structure.

If A is an abelian group containing an element x of finite order n then $nx = 0$. Thus, in contrast to vector spaces, a $\mathbb{Z}$ -module may have nonzero elements x such that $nx = 0$ for some nonzero ring element n . In particular, if A has order m , then by Lagrange’s Theorem (Corollary 9, Section 3.2) $mx = 0$, for all $x \in A$. Note that then A is a module over $\mathbb{Z}/m\mathbb{Z}$.

In particular, if p is a prime and A is an abelian group (written additively) such that $px = 0$, for all $x \in A$, then (as noted in Example 5) A is a $\mathbb{Z}/p\mathbb{Z}$ -module, i.e., can be considered as a vector space over the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. For instance, the Klein 4-group is a (2-dimensional) vector space over $\mathbb{F}_2$. These groups are the *elementary abelian p -groups* discussed in Section 4.4 (see, in particular, Proposition 17(3)).

The next example is also of fundamental importance and will form the basis for our study of canonical forms of matrices in Sections 12.2 and 12.3.

Example: ($F[x]$ -modules)

Let F be a field, let x be an indeterminate and let R be the polynomial ring $F[x]$. Let V be a vector space over F and let T be a linear transformation from V to V (we shall review the theory of linear transformations in the next chapter — for the purposes of this example one only needs to know the definition of a linear transformation). We have already seen that V is an F -module; the linear map T will enable us to make V into an $F[x]$ -module.

First, for the nonnegative integer n , define

$$\begin{aligned} T^0 &= I, \\ &\vdots \\ T^n &= T \circ T \circ \cdots \circ T \quad (n \text{ times}) \end{aligned}$$

where I is the identity map from V to V and $\circ$ denotes function composition (which makes sense because the domain and codomain of T are the same). Also, for any two linear transformations A, B from V to V and elements $\alpha, \beta \in F$, let $\alpha A + \beta B$ be defined by

$$(\alpha A + \beta B)(v) = \alpha(A(v)) + \beta(B(v))$$

(i.e., addition and scalar multiplication of linear transformations are defined pointwise). Then $\alpha A + \beta B$ is easily seen to be a linear transformation from V to V , so that linear combinations of linear transformations are again linear transformations.

We now define the action of any polynomial in x on V . Let $p(x)$ be the polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_0, \dots, a_n \in F$. For each $v \in V$ define an action of the ring element $p(x)$ on the module element v by

$$\begin{aligned} p(x)v &= (a_n T^n + a_{n-1} T^{n-1} + \cdots + a_1 T + a_0)(v) \\ &= a_n T^n(v) + a_{n-1} T^{n-1}(v) + \cdots + a_1 T(v) + a_0 v \end{aligned}$$

(i.e., $p(x)$ acts by substituting the linear transformation T for x in $p(x)$ and applying the resulting linear transformation to v). Put another way, x acts on V as the linear transformation T and we extend this to an action of all of $F[x]$ on V in a natural way. It is easy to check that this definition of an action of $F[x]$ on V satisfies all the module axioms and makes V into an $F[x]$ -module.

The field F is naturally a subring of $F[x]$ (the constant polynomials) and the action of these field elements is by definition the same as their action when viewed as constant polynomials. In other words, the definition of the $F[x]$ action on V is consistent with the given action of the field F on the vector space V , i.e., the definition *extends* the action of F to an action of the larger ring $F[x]$.

The way $F[x]$ acts on V depends on the choice of T so that there are in general many different $F[x]$ -module structures on the same vector space V . For instance, if $T = 0$, and $p(x), v$ are as above, then $p(x)v = a_0 v$, that is, the polynomial $p(x)$ acts on v simply by multiplying by the constant term of $p(x)$, so that the $F[x]$ -module structure is just the F -module structure. If, on the other hand, T is the identity transformation (so $T^n(v) = v$, for all n and v), then $p(x)v = a_n v + a_{n-1} v + \cdots + a_0 v = (a_n + \cdots + a_0)v$, so that now $p(x)$ multiplies v by the sum of the coefficients of $p(x)$.

To give another specific example, let V be affine n -space F^n and let T be the “shift operator”

$$T(x_1, x_2, \dots, x_n) = (x_2, x_3, \dots, x_n, 0).$$

Let e_i be the usual i^{th} basis vector $(0, 0, \dots, 0, 1, 0, \dots, 0)$ where the 1 is in position i . Then

$$T^k(e_i) = \begin{cases} e_{i-k} & \text{if } i > k \\ 0 & \text{if } i \leq k \end{cases}$$

so for example, if $m < n$,

$$(a_m x^m + a_{m-1} x^{m-1} + \dots + a_0) e_n = (0, \dots, 0, a_m, a_{m-1}, \dots, a_0).$$

From this we can determine the action of any polynomial on any vector.

The construction of an $F[x]$ -module from a vector space V over F and a linear transformation T from V to V in fact describes *all* $F[x]$ -modules; namely, an $F[x]$ -module is a vector space together with a linear transformation which specifies the action of x . This is because if V is any $F[x]$ -module, then V is an F -module and the action of the ring element x on V is a linear transformation from V to V . The axioms for a module ensure that the actions of F and x on V uniquely determine the action of any element of $F[x]$ on V . Thus there is a bijection between the collection of $F[x]$ -modules and the collection of pairs V, T

$$\left\{ \begin{array}{l} V \text{ an } F[x]\text{-module} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} V \text{ a vector space over } F \\ \text{and} \\ T : V \rightarrow V \text{ a linear transformation} \end{array} \right\}$$

given by

the element x acts on V as the linear transformation T .

Now we consider $F[x]$ -submodules of V where, as above, V is any $F[x]$ -module and T is the linear transformation from V to V given by the action of x . An $F[x]$ -submodule W of V must first be an F -submodule, i.e., W must be a vector subspace of V . Secondly, W must be sent to itself under the action of the ring element x , i.e., we must have $T(w) \in W$, for all $w \in W$. Any vector subspace U of V such that $T(U) \subseteq U$ is called *T -stable* or *T -invariant*. If U is any T -stable subspace of V it follows that $T^n(U) \subseteq U$, for all $n \in \mathbb{Z}^+$ (for example, $T(U) \subseteq U$ implies $T^2(U) = T(T(U)) \subseteq T(U) \subseteq U$). Moreover any linear combination of powers of T then sends U into U so that U is also stable by the action of any polynomial in T . Thus U is an $F[x]$ -submodule of V . This shows that

the $F[x]$ -submodules of V are precisely the T -stable subspaces of V .

In terms of the bijection above,

$$\left\{ \begin{array}{l} W \text{ an } F[x]\text{-submodule} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} W \text{ a subspace of } V \\ \text{and} \\ W \text{ is } T\text{-stable} \end{array} \right\}$$

which gives a complete dictionary between $F[x]$ -modules V and vector spaces V together with a given linear transformation T from V to V .

For instance, if T is the shift operator defined on affine n -space above and k is any integer in the range $0 \leq k \leq n$, then the subspace

$$U_k = \{ (x_1, x_2, \dots, x_k, 0, \dots, 0) \mid x_i \in F \}$$

is clearly T -stable so is an $F[x]$ -submodule of V .

We emphasize that an abelian group M may have many different R -module structures, even if the ring R does not vary (in the same way that a given group G may act in many ways as a permutation group on some fixed set Ω). We shall see that the structure of an R -module is reflected by the ideal structure of R . When R is a field (the subject of the next chapter) all R -modules will be seen to be products of copies of R (as in Example 3 above).

We shall see in Chapter 12 that the relatively simple ideal structure of the ring $F[x]$ (recall that $F[x]$ is a Principal Ideal Domain) forces the $F[x]$ -module structure of V to be correspondingly uncomplicated, and this in turn provides a great deal of information about the linear transformation T (in particular, gives some nice matrix representations for T : its *rational canonical form* and its *Jordan canonical form*). Moreover, the same arguments which *classify* finitely generated $F[x]$ -modules apply to any Principal Ideal Domain R , and when these are invoked for $R = \mathbb{Z}$, we obtain the Fundamental Theorem of Finitely Generated Abelian Groups. These results generalize the theorem that every finite dimensional vector space has a basis.

In Part VI of the book we shall study modules over certain noncommutative rings (group rings) and see that this theory in some sense generalizes both the study of $F[x]$ -modules in Chapter 12 and the notion of a permutation representation of a finite group.

We establish a submodule criterion analogous to that for subgroups of a group in Section 2.1.

Proposition 1. (*The Submodule Criterion*) Let R be a ring and let M be an R -module. A subset N of M is a submodule of M if and only if

- (1) $N \neq \emptyset$, and
- (2) $x + ry \in N$ for all $r \in R$ and for all $x, y \in N$.

Proof: If N is a submodule, then $0 \in N$ so $N \neq \emptyset$. Also N is closed under addition and is sent to itself under the action of elements of R . Conversely, suppose (1) and (2) hold. Let $r = -1$ and apply the subgroup criterion (in additive form) to see that N is a subgroup of M . In particular, $0 \in N$. Now let $x = 0$ and apply hypothesis (2) to see that N is sent to itself under the action of R . This establishes the proposition.

We end this section with an important definition and some examples.

Definition. Let R be a commutative ring with identity. An R -algebra is a ring A with identity together with a ring homomorphism $f : R \rightarrow A$ mapping 1_R to 1_A such that the subring $f(R)$ of A is contained in the center of A .

If A is an R -algebra then it is easy to check that A has a natural left and right (unital) R -module structure defined by $r \cdot a = a \cdot r = f(r)a$ where $f(r)a$ is just the multiplication in the ring A (and this is the same as $af(r)$ since by assumption $f(r)$ lies in the center of A). In general it is possible for an R -algebra A to have other left (or right) R -module structures, but unless otherwise stated, this natural module structure on an algebra will be assumed.

Definition. If A and B are two R -algebras, an R -algebra homomorphism (or isomorphism) is a ring homomorphism (isomorphism, respectively) $\varphi : A \rightarrow B$ mapping 1_A to 1_B such that $\varphi(r \cdot a) = r \cdot \varphi(a)$ for all $r \in R$ and $a \in A$.

Examples

Let R be a commutative ring with 1.

- (1) Any ring with identity is a $\mathbb{Z}$ -algebra.
- (2) For any ring A with identity, if R is a subring of the center of A containing the identity of A then A is an R -algebra. In particular, a commutative ring A containing 1 is an R -algebra for any subring R of A containing 1. For example, the polynomial ring $R[x]$ is an R -algebra, the polynomial ring over R in any number of variables is an R -algebra, and the group ring RG for a finite group G is an R -algebra (cf. Section 7.2).
- (3) If A is an R -algebra then the R -module structure of A depends only on the subring $f(R)$ contained in the center of A as in the previous example. If we replace R by its image $f(R)$ we see that “up to a ring homomorphism” every algebra A arises from a subring of the center of A that contains 1_A .
- (4) A special case of the previous example occurs when $R = F$ is a field. In this case F is isomorphic to its image under f , so we can identify F itself as a subring of A . Hence, saying that A is an algebra over a field F is the same as saying that the ring A contains the field F in its center and the identity of A and of F are the same (this last condition is necessary, cf. Exercise 23).

Suppose that A is an R -algebra. Then A is a ring with identity that is a (unital) left R -module satisfying $r \cdot (ab) = (r \cdot a)b = a(r \cdot b)$ for all $r \in R$ and $a, b \in A$ (these are all equal to the product $f(r)ab$ in the ring A —recall that $f(R)$ is contained in the center of A). Conversely, these conditions on a ring A define an R -algebra, and are sometimes used as the definition of an R -algebra (cf. Exercise 22).

EXERCISES

In these exercises R is a ring with 1 and M is a left R -module.

1. Prove that $0m = 0$ and $(-1)m = -m$ for all $m \in M$.
2. Prove that $R^\times$ and M satisfy the two axioms in Section 1.7 for a group action of the multiplicative group $R^\times$ on the set M .
3. Assume that $rm = 0$ for some $r \in R$ and some $m \in M$ with $m \neq 0$. Prove that r does not have a left inverse (i.e., there is no $s \in R$ such that $sr = 1$).
4. Let M be the module R^n described in Example 3 and let $I_1, I_2, \dots, I_n$ be left ideals of R . Prove that the following are submodules of M :
 - (a) $\{(x_1, x_2, \dots, x_n) \mid x_i \in I_i\}$
 - (b) $\{(x_1, x_2, \dots, x_n) \mid x_i \in R \text{ and } x_1 + x_2 + \dots + x_n = 0\}$.
5. For any left ideal I of R define

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I, m_i \in M \right\}$$

to be the collection of all finite sums of elements of the form am where $a \in I$ and $m \in M$. Prove that IM is a submodule of M .

6. Show that the intersection of any nonempty collection of submodules of an R -module is a submodule.

7. Let $N_1 \subseteq N_2 \subseteq \cdots$ be an ascending chain of submodules of M . Prove that $\bigcup_{i=1}^{\infty} N_i$ is a submodule of M .
8. An element m of the R -module M is called a *torsion element* if $rm = 0$ for some nonzero element $r \in R$. The set of torsion elements is denoted

$$\text{Tor}(M) = \{m \in M \mid rm = 0 \text{ for some nonzero } r \in R\}.$$

- (a) Prove that if R is an integral domain then $\text{Tor}(M)$ is a submodule of M (called the *torsion submodule* of M).
- (b) Give an example of a ring R and an R -module M such that $\text{Tor}(M)$ is not a submodule. [Consider the torsion elements in the R -module R .]
- (c) If R has zero divisors show that every nonzero R -module has nonzero torsion elements.
9. If N is a submodule of M , the *annihilator of N in R* is defined to be $\{r \in R \mid rn = 0 \text{ for all } n \in N\}$. Prove that the annihilator of N in R is a 2-sided ideal of R .
10. If I is a right ideal of R , the *annihilator of I in M* is defined to be $\{m \in M \mid am = 0 \text{ for all } a \in I\}$. Prove that the annihilator of I in M is a submodule of M .
11. Let M be the abelian group (i.e., $\mathbb{Z}$ -module) $\mathbb{Z}/24\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/50\mathbb{Z}$.
 - (a) Find the annihilator of M in $\mathbb{Z}$ (i.e., a generator for this principal ideal).
 - (b) Let $I = 2\mathbb{Z}$. Describe the annihilator of I in M as a direct product of cyclic groups.
12. In the notation of the preceding exercises prove the following facts about annihilators.
 - (a) Let N be a submodule of M and let I be its annihilator in R . Prove that the annihilator of I in M contains N . Give an example where the annihilator of I in M does not equal N .
 - (b) Let I be a right ideal of R and let N be its annihilator in M . Prove that the annihilator of N in R contains I . Give an example where the annihilator of N in R does not equal I .
13. Let I be an ideal of R . Let M' be the subset of elements a of M that are annihilated by some power, I^k , of the ideal I , where the power may depend on a . Prove that M' is a submodule of M . [Use Exercise 7.]
14. Let z be an element of the center of R , i.e., $zr = rz$ for all $r \in R$. Prove that zM is a submodule of M , where $zM = \{zm \mid m \in M\}$. Show that if R is the ring of 2×2 matrices over a field and e is the matrix with a 1 in position 1,1 and zeros elsewhere then eR is *not* a left R -submodule (where $M = R$ is considered as a left R -module as in Example 1) — in this case the matrix e is not in the center of R .
15. If M is a finite abelian group then M is naturally a $\mathbb{Z}$ -module. Can this action be extended to make M into a $\mathbb{Q}$ -module?
16. Prove that the submodules U_k described in the example of $F[x]$ -modules are all of the $F[x]$ -submodules for the shift operator.
17. Let T be the shift operator on the vector space V and let $e_1, \dots, e_n$ be the usual basis vectors described in the example of $F[x]$ -modules. If $m \geq n$ find $(a_m x^m + a_{m-1} x^{m-1} + \cdots + a_0) e_n$.
18. Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let T be the linear transformation from V to V which is rotation clockwise about the origin by $\pi/2$ radians. Show that V and 0 are the only $F[x]$ -submodules for this T .
19. Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let T be the linear transformation from V to V which is projection onto the y -axis. Show that V , 0 , the x -axis and the y -axis are the only $F[x]$ -submodules for this T .
20. Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let T be the linear transformation from V to V which is rotation clockwise about the origin by π radians. Show that *every* subspace of V is an

$F[x]$ -submodule for this T .

21. Let $n \in \mathbb{Z}^+$, $n > 1$ and let R be the ring of $n \times n$ matrices with entries from a field F . Let M be the set of $n \times n$ matrices with arbitrary elements of F in the first column and zeros elsewhere. Show that M is a submodule of R when R is considered as a left module over itself, but M is not a submodule of R when R is considered as a right R -module.
22. Suppose that A is a ring with identity 1_A that is a (unital) left R -module satisfying $r \cdot (ab) = (r \cdot a)b = a(r \cdot b)$ for all $r \in R$ and $a, b \in A$. Prove that the map $f : R \rightarrow A$ defined by $f(r) = r \cdot 1_A$ is a ring homomorphism mapping 1_R to 1_A and that $f(R)$ is contained in the center of A . Conclude that A is an R -algebra and that the R -module structure on A induced by its algebra structure is precisely the original R -module structure.
23. Let A be the direct product ring $\mathbb{C} \times \mathbb{C}$ (cf. Section 7.6). Let τ_1 denote the identity map on $\mathbb{C}$ and let τ_2 denote complex conjugation. For any pair $p, q \in \{1, 2\}$ (not necessarily distinct) define

$$f_{p,q} : \mathbb{C} \rightarrow \mathbb{C} \times \mathbb{C} \quad \text{by} \quad f_{p,q}(z) = (\tau_p(z), \tau_q(z)).$$

So, for example, $f_{2,1} : z \mapsto (\bar{z}, z)$, where $\bar{z}$ is the complex conjugate of z , i.e., $\tau_2(z)$.

- (a) Prove that each $f_{p,q}$ is an injective ring homomorphism, and that they all agree on the subfield $\mathbb{R}$ of $\mathbb{C}$. Deduce that A has four distinct $\mathbb{C}$ -algebra structures. Explicitly give the action $z \cdot (u, v)$ of a complex number z on an ordered pair in A in each case.
- (b) Prove that if $f_{p,q} \neq f_{p',q'}$ then the identity map on A is *not* a $\mathbb{C}$ -algebra homomorphism from A considered as a $\mathbb{C}$ -algebra via $f_{p,q}$ to A considered as a $\mathbb{C}$ -algebra via $f_{p',q'}$ (although the identity is an $\mathbb{R}$ -algebra isomorphism).
- (c) Prove that for any pair p, q there is some ring isomorphism from A to itself such that A is isomorphic as a $\mathbb{C}$ -algebra via $f_{p,q}$ to A considered as $\mathbb{C}$ -algebra via $f_{1,1}$ (the “natural” $\mathbb{C}$ -algebra structure on A).

Remark: In the preceding exercise $A = \mathbb{C} \times \mathbb{C}$ is not a $\mathbb{C}$ -algebra over either of the direct factor component copies of $\mathbb{C}$ (for example the subring $\mathbb{C} \times 0 \cong \mathbb{C}$) since it is not a unital module over these copies of $\mathbb{C}$ (the 1 of these subrings is not the same as the 1 of A).

10.2 QUOTIENT MODULES AND MODULE HOMOMORPHISMS

This section contains the basic theory of quotient modules and module homomorphisms.

Definition. Let R be a ring and let M and N be R -modules.

- (1) A map $\varphi : M \rightarrow N$ is an *R -module homomorphism* if it respects the R -module structures of M and N , i.e.,
 - (a) $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in M$ and
 - (b) $\varphi(rx) = r\varphi(x)$, for all $r \in R, x \in M$.
- (2) An R -module homomorphism is an *isomorphism (of R -modules)* if it is both injective and surjective. The modules M and N are said to be *isomorphic*, denoted $M \cong N$, if there is some R -module isomorphism $\varphi : M \rightarrow N$.
- (3) If $\varphi : M \rightarrow N$ is an R -module homomorphism, let $\ker \varphi = \{m \in M \mid \varphi(m) = 0\}$ (the *kernel* of φ) and let $\varphi(M) = \{n \in N \mid n = \varphi(m) \text{ for some } m \in M\}$ (the *image* of φ , as usual).
- (4) Let M and N be R -modules and define $\text{Hom}_R(M, N)$ to be the set of all R -module homomorphisms from M into N .

Any R -module homomorphism is also a homomorphism of the additive groups, but not every group homomorphism need be a module homomorphism (because condition (b) may not be satisfied). The unqualified term “isomorphism” when applied to R -modules will always mean R -module isomorphism. When the symbol $\cong$ is used without qualification it will denote an isomorphism of the respective structures (which will be evident from the context).

It is an easy exercise using the submodule criterion (Proposition 1) to show that kernels and images of R -module homomorphisms are submodules.

Examples

- (1) If R is a ring and $M = R$ is a module over itself, then R -module homomorphisms (even from R to itself) need not be ring homomorphisms and ring homomorphisms need not be R -module homomorphisms. For example, when $R = \mathbb{Z}$ the $\mathbb{Z}$ -module homomorphism $x \mapsto 2x$ is not a ring homomorphism (1 does not map to 1). When $R = F[x]$ the ring homomorphism $\varphi : f(x) \mapsto f(x^2)$ is not an $F[x]$ -module homomorphism (if it were, we would have $x^2 = \varphi(x) = \varphi(x \cdot 1) = x\varphi(1) = x$).
- (2) Let R be a ring, let $n \in \mathbb{Z}^+$ and let $M = R^n$. One easily checks that for each $i \in \{1, \dots, n\}$ the projection map

$$\pi_i : R^n \rightarrow R \quad \text{by} \quad \pi_i(x_1, \dots, x_n) = x_i$$

is a surjective R -module homomorphism with kernel equal to the submodule of n -tuples which have a zero in position i .

- (3) If R is a field, R -module homomorphisms are called *linear transformations*. These will be studied extensively in Chapter 11.
- (4) For the ring $R = \mathbb{Z}$ the action of ring elements (integers) on any $\mathbb{Z}$ -module amounts to just adding and subtracting within the (additive) abelian group structure of the module so that in this case condition (b) of a homomorphism is implied by condition (a). For example, $\varphi(2x) = \varphi(x + x) = \varphi(x) + \varphi(x) = 2\varphi(x)$, etc. It follows that

$\mathbb{Z}$ -module homomorphisms are the same as abelian group homomorphisms.

- (5) Let R be a ring, let I be a 2-sided ideal of R and suppose M and N are R -modules annihilated by I (i.e., $am = 0$ and $an = 0$ for all $a \in I$, $n \in N$ and $m \in M$). Any R -module homomorphism from N to M is then automatically a homomorphism of (R/I) -modules (see Example 5 of Section 1). In particular, if A is an additive abelian group such that for some prime p , $px = 0$ for all $x \in A$, then any group homomorphism from A to itself is a $\mathbb{Z}/p\mathbb{Z}$ -module homomorphism, i.e., is a linear transformation over the field $\mathbb{F}_p$. In particular, the group of all (group) automorphisms of A is the group of invertible linear transformations from A to itself: $GL(A)$.

Proposition 2. Let M , N and L be R -modules.

- (1) A map $\varphi : M \rightarrow N$ is an R -module homomorphism if and only if $\varphi(rx + y) = r\varphi(x) + \varphi(y)$ for all $x, y \in M$ and all $r \in R$.
- (2) Let φ, ψ be elements of $\text{Hom}_R(M, N)$. Define $\varphi + \psi$ by

$$(\varphi + \psi)(m) = \varphi(m) + \psi(m) \quad \text{for all } m \in M.$$

Then $\varphi + \psi \in \text{Hom}_R(M, N)$ and with this operation $\text{Hom}_R(M, N)$ is an abelian group. If R is a commutative ring then for $r \in R$ define $r\varphi$ by

$$(r\varphi)(m) = r(\varphi(m)) \quad \text{for all } m \in M.$$